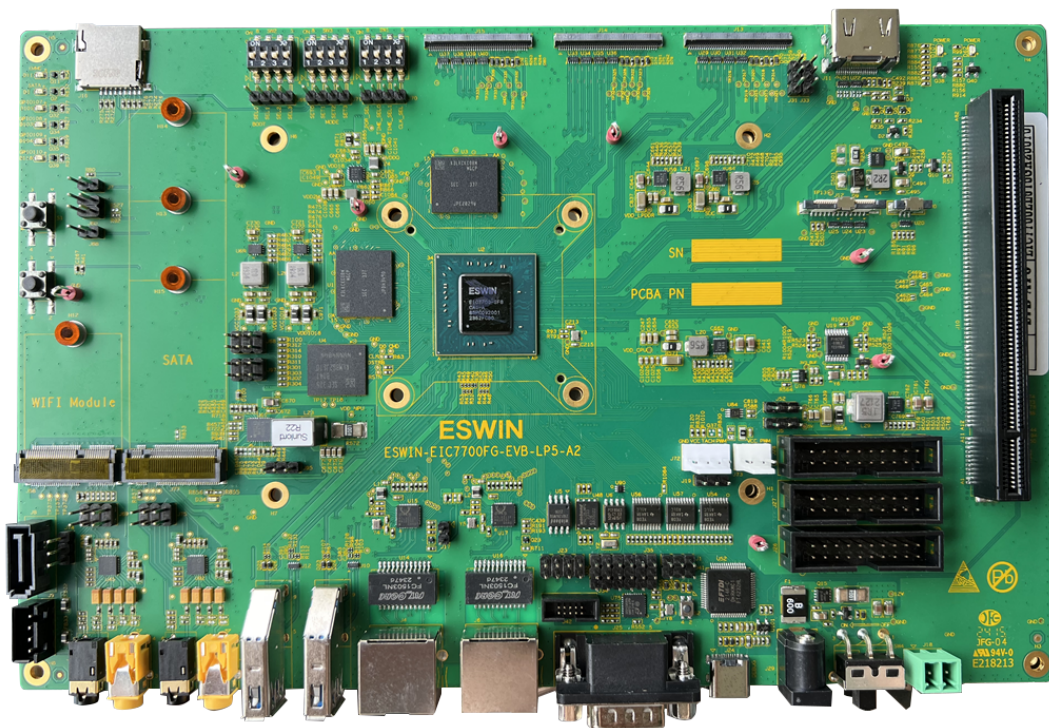


ESWIN

EIC7700FG EVB LP5 A2 User Manual

Rev 2.0

2026-3-26



Notice

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Change History

Version	Date	Description
Rev 2.0	2026/03/26	Restructured document layout
Rev 1.2	2024/09/12	Updated flashing section
Rev 1.1	2024/08/20	Errata: Corrected resistor configuration diagrams for PCIE RC and EP modes
Rev 1.0	2024/08/10	Updated software and image flashing methods based on the “Development Board Image Installation and Upgrade Manual” to make the complete EVB user documentation more comprehensive
Rev 0.6	2024/07/01	Modified based on user feedback and test results. Integrated embedded firmware content, such as basic firmware upgrade and flashing to ensure effective hardware operation, into the hardware documentation
Rev 0.4	2024/06/01	Initial version

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1. Safety Notices

1.1. Copyright and Usage Statement

The EIC7700FG_EVB_LP5_A2 development board (hereinafter referred to as “EVB-A2” or “Product”) provided by ESWIN is intended for end-users only, and solely for testing and evaluating ESWIN’ s semiconductor components and software applications (i.e., “Intended Use”). No person shall resell, redistribute, transfer the Product, or otherwise profit from it.

Please note that this document is subject to periodic updates. You may contact ESWIN Sales and After-sales Support to obtain the latest version.

1.2. Important Safety Information

1.2.1. Warnings

- Any power supply used with the EVB-A2 must comply with the laws, regulations and standards of the country/region where it is used.
- The EVB-A2 must be operated in a well-ventilated environment.
- This product should only be used on a flat, non-conductive surface, and must never come into contact with conductors at any time.
- Connecting incompatible devices to the EVB-A2 may damage the development board.
- All accessories and peripheral devices used with the EVB-A2 must comply with the standards of the country/region of use and bear the appropriate certification labels to ensure safety requirements are met.
- All cables and connection equipment used with the EVB-A2 must be properly insulated according to safety standards and requirements.

1.2.2. Safety Instructions

To avoid failure or damage to the EVB-A2, please comply with the following requirements:

- Do not expose the EVB-A2 to water or moisture.
- During operation, do not place or position the EVB-A2 near conductive surfaces.
- Do not expose the EVB-A2 to high temperatures; use the EVB-A2 only at room temperature in a well-ventilated environment as specified in this manual.
- Avoid any mechanical or electrical damage to the printed circuit board, interfaces and integrated circuits on the EVB-A2.
- Do not touch, pick up or move the EVB-A2 while it is powered on.

2. Introduction

The EIC7700FG_EVB_LP5_A2 (hereinafter referred to as EVB-A2) is a high-performance, full-featured development and evaluation board based on ESWIN's EIC7700 AI SoC. Integrating a flagship-grade RISC-V CPU, a powerful AI acceleration unit, high-speed LPDDR5 memory, and a rich array of industrial-grade peripheral interfaces, this platform serves as an ideal hardware solution for prototype development, algorithm verification, and performance evaluation in fields such as edge AI computing, machine vision, high-end embedded systems, and network applications.

2.1. Hardware Features

The EVB-A2 delivers top-tier hardware configurations within a standard form factor. Detailed specifications are provided in the table below:

Table 1: EIC7700FG_EVB_LP5_A2 Key Hardware Specifications

Parameter	Description
Form Factor	150.04mm x 235mm
CPU	64-bit Out-of-order execution RISC-V Quad-core Processor
AI Processor	Up to 13.3 TOPS, supporting INT8/INT16/FP16 precision
Memory	On-board 16GB/32GB LPDDR5, 6400Mbps data rate
Storage	1x On-board 32GB eMMC
	1x On-board 16MB SPI Flash
	1x M.2 Key B&M Connector (for SSD)
	1x Micro SD Card Slot
Network	2x Gigabit Ethernet RJ45 Ports
	1x M.2 Key E Connector (for Wi-Fi/BT Module)
Display	1x HDMI 2.0 Type-A Port, supporting 4K@60fps
	1x MIPI DSI 4-lane FPC Connector
Camera	3x MIPI CSI 2+2 lane or 4-lane FPC Connectors
Audio	2x 3.5mm Audio Output Jacks
	2x 3.5mm CTIA Standard Headset Jacks
PCIe	1x PCI Express Gen3 x4 lane (via PCIe X16 Slot)
USB	2x Standard USB 3.2 Gen1 Type-A Ports
Fan	1x 2.54mm 4-pin 12V Fan Header
	1x 2.54mm 3-pin 5V Fan Header
Buttons	1x Reset Button
	1x Boot Select Button
Power Supply	1x DC 2.5mm Jack (12V, 5A)
	1x Power Slide Switch (On/Off)
	1x Remote Power On/Off Connector
Debug	1x USB Type-C Port (for Debug UART and JTAG)
Operating Temperature	-20~60 °C

2.2. Functional Layout

To help you quickly identify key components on the board, please refer to the figure below:

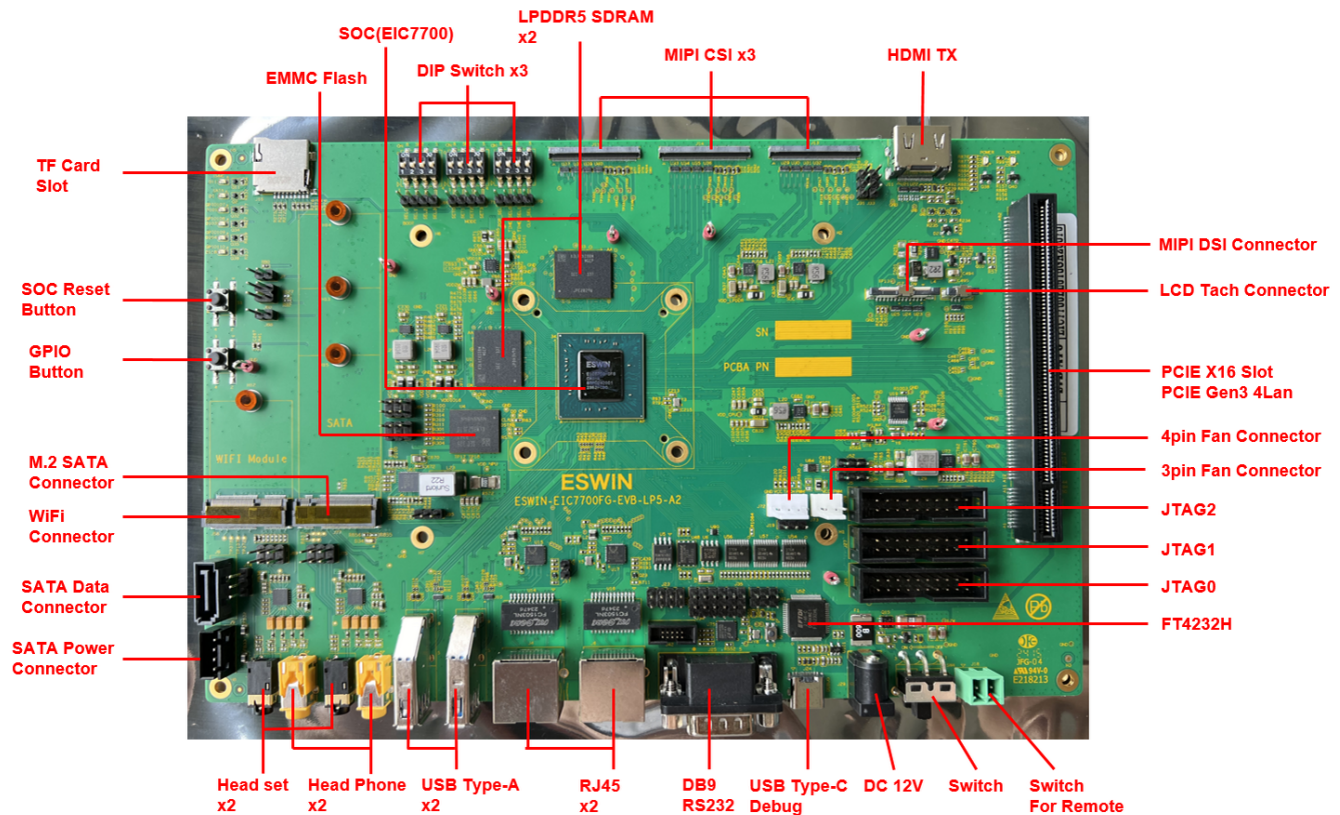


Figure 1: EVB-A2 Functional Layout Diagram

2.3. Accessories List

To ensure you can successfully unbox and use the EVB-A2 development board, the standard product package includes the following essential accessories. Upon receiving the product, please verify against this list.

Item	Accessory Name	Quantity	Remarks
1	EIC7700FG EVB-A2 Development Board	1 Set	Pre-installed with heatsink and fan
2	Power Adapter (12V/5A)	1 pc	Input: 100-240V~50/60Hz, DC2.5mm connector
3	USB Type-C Cable	1 pc	For connecting debug interface
4	Screws and Standoffs	4 pcs each	For mounting and securing the development board

2.4. Supporting Software and Documentation

The EVB-A2 provides a complete software and hardware support ecosystem. In addition to this *User Manual* which guides hardware usage and basic flashing, for deeper system customization, driver development, and AI application deployment, please refer to the following resources:

- [BSP Quick Start Guide](#): Guides you on how to set up a cross-compilation environment, customize the Linux kernel, build root filesystems, and compile the complete SDK.
- [ESSDK Quick Start Guide](#): Details the structure and usage of the ESWIN SDK, as well as how to perform AI model quantization, compilation, and on-board inference deployment.

You can obtain the latest documentation, tools, and images through ESWIN' s official technical support channels.

3. Design Block Diagrams

3.1. EIC7700 Chip Block Diagram

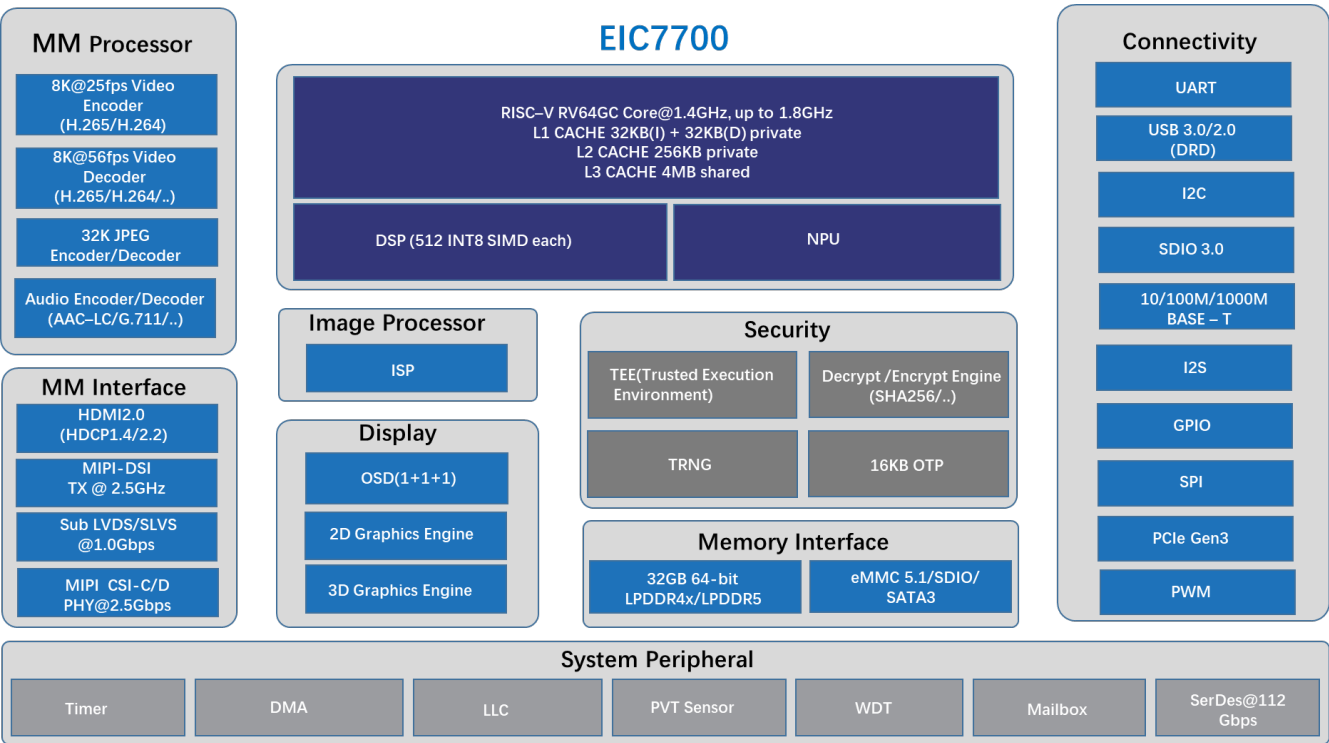


Figure 2: EIC7700 Chip Block Diagram

3.2. System Block Diagram

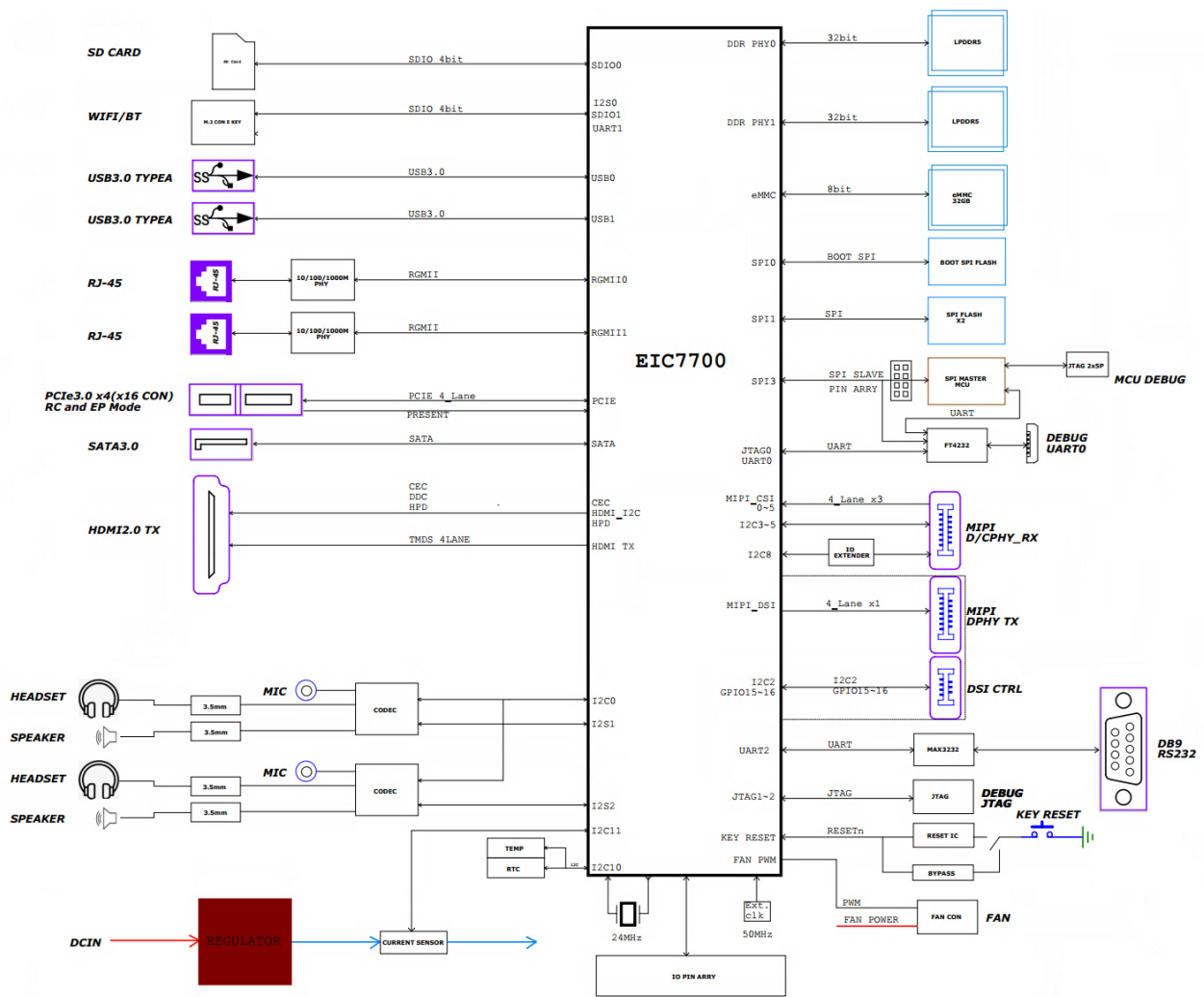


Figure 3: System Block Diagram

4. Hardware Introduction and Usage Guide

This chapter details the hardware features, interface definitions, key configurations, and basic operation methods of the EVB-A2 development board. It serves as the foundation for using and evaluating this development platform.

4.1. Preparation

Before powering on the development board, please complete all preparation steps in this section to ensure device safety and proper operation.

4.1.1. Heatsink and Fan

To prevent overheating and damage to the core chip, the EVB-A2 comes pre-installed with a heatsink and a connected 5V fan.

Warning

Do not power on the development board without the heatsink or fan correctly installed and connected. Failure to do so may cause permanent damage to the EIC7700 SoC due to overheating. Please ensure the heatsink and fan (connector J73) are securely connected before operation.

4.1.2. Connecting Peripherals

To facilitate subsequent debugging, operation, and functional verification, it is recommended to connect the following common peripherals before the first power-on:

Peripheral	Connection Interface	Main Purpose & Description
Power Supply	DC 12V Power Jack (DJ1)	Connect the included 12V/5A DC power adapter. This is the main power supply method for the board.
Debugging & CLI	USB Type-C Debug Port	Connect to a PC for serial debugging, viewing system boot messages, and console login. Driver installation is required for first-time use on Windows.
Display Output	HDMI 2.0 Port	Connect a monitor to view the graphical desktop or system boot interface.
Input Device	Any USB 3.2 Type-A Port	Connect a keyboard and mouse to operate the GUI or command line after boot.
Network	Any Gigabit Ethernet (RJ45) Port	Connect the Ethernet cable to connect the development board to the local area network, facilitating network installation, updates, and remote access.

4.1.3. Power On

The EVB-A2 is powered by a 12V/5A DC power adapter.

Steps:

1. Insert the DC plug of the power adapter into the DJ1 connector on the board.
2. Locate the Power Switch on the board and slide it to the **ON** position.
3. Observe the board: the fan should start spinning, and the on-board LED should light up. Simultaneously, system boot logs should appear on the serial terminal, indicating a successful startup.

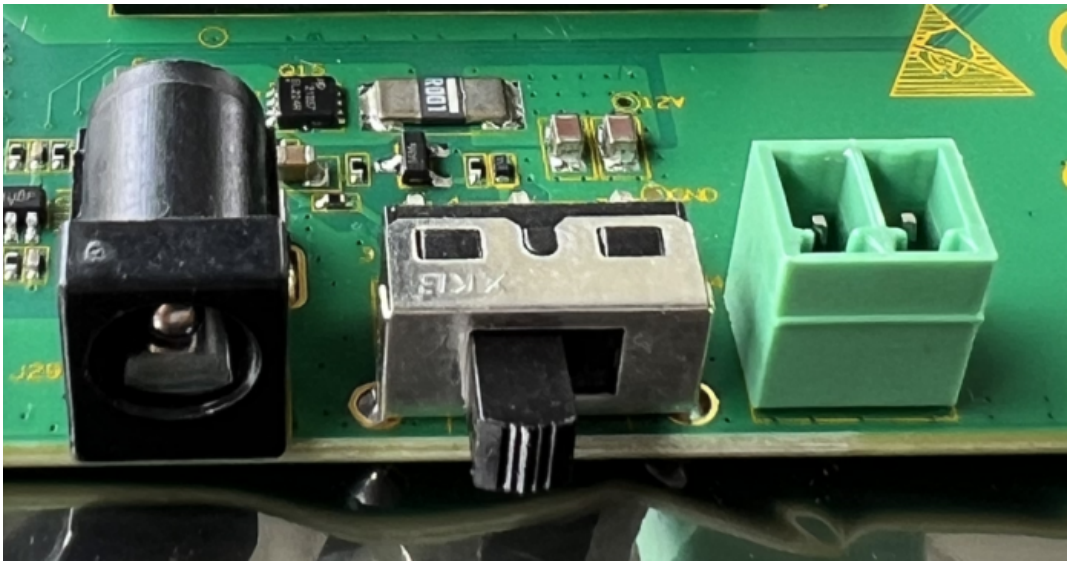


Figure 5: Power Socket and Switch

4.2. Power On/Off, Boot Modes, and Status Identification

4.2.1. Button Functions and Operation

The EVB-A2 provides two core user buttons:

Button Name	Main Function	Operation Method
Power Switch	Controls main board power on/off.	Physical slide switch. Slide to ON for power, OFF for cut-off. This is the fundamental power control.
SoC Reset Button (S1)	Hardware resets the EIC7700 SoC main system.	Short press: Immediately triggers a hardware restart of the SoC, equivalent to a PC’s “Reset” button.

4.2.2. Boot Mode Selection

The EVB-A2 uses the on-board 4-position DIP switch SW2 to select the Bootloader source. This is a core operation for system flashing and recovery.

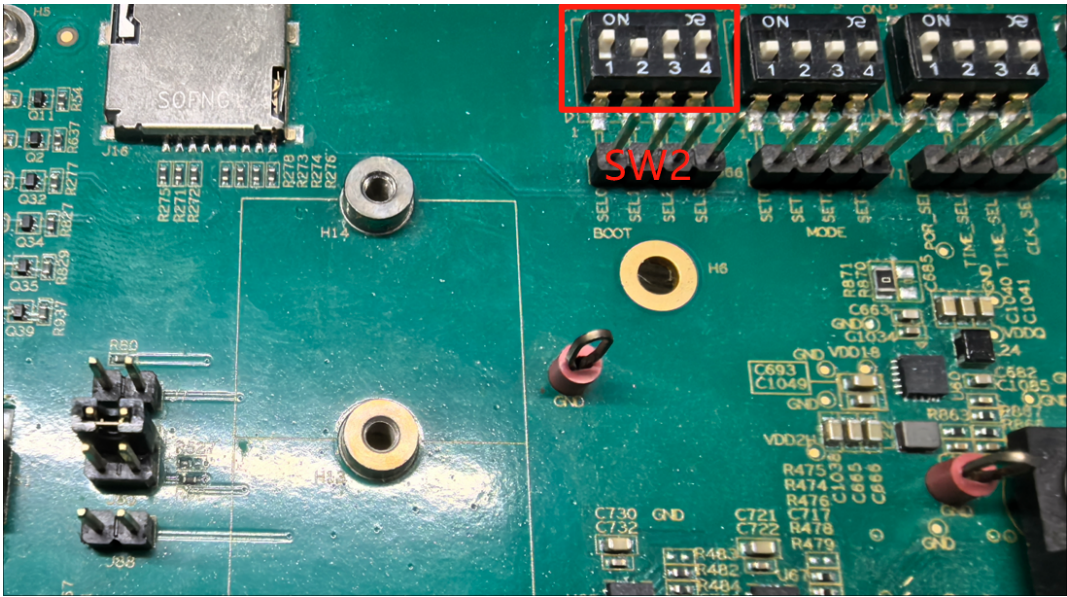


Figure 6: DIP Switch SW2

- **Switch Description:** The side marked “ON” represents 0, and the side with numbers “1,2,3,4” represents 1.
- **Common Settings:** The following are the two most critical boot mode settings:

DIP Status (SW2.1 - SW2.4)	Boot Mode	Applicable Scenario & Operation Guide
0100	Boot from SPI Flash	Normal Usage Mode. Boots the installed U-Boot and OS from the on-board SPI Flash. Keep this mode for daily development.
1100	Boot from USB (Device Mode)	Forced Recovery Mode. Used for forced recovery when the Bootloader in SPI Flash is corrupted and cannot boot. In this mode, the board will be recognized as a USB drive by the PC, allowing recovery images to be copied.

Tip

1. **Power Off Operation:** Before changing DIP switch settings, you must completely power off the board (turn off Power Switch or unplug power).
2. **Set and Power On:** Set the SW2 DIP switches to the correct position as per the table above, then re-power on for the new settings to take effect.

4.2.3. Power On/Off Procedures

Combining the power switch and boot modes, the standard workflow is as follows:

1. **Power On:**
 - Ensure SW2 is set to 0100 (Boot from SPI Flash).
 - Connect the power adapter and slide the Power Switch to ON.
 - View system boot progress via the serial terminal or HDMI monitor.
2. **Normal Shutdown:**
 - Execute the shutdown command in the running OS.
 - Wait for the system to fully shut down, then slide the Power Switch to OFF.

3. Forced Power Off:

- Use only when the system is completely unresponsive.
- Slide the Power Switch directly to OFF.
- **Note:** This operation may cause data loss.

4.2.4. Serial Debugging

The EVB-A2 provides UART0 serial debugging via the USB Type-C debug port, which also multiplexes JTAG and SPI functions.

1. **Hardware Connection:** Use the included USB Type-C cable to connect the board's debug port to the PC.
2. **Driver Installation (Windows):** For first-time use on Windows, install the virtual serial port driver so the PC can recognize the COM port. See [Windows Driver Installation and Debugging Guide](#) for details.
3. **Software Configuration:** Use a serial terminal tool on the PC (e.g., MobaXterm, Putty, SecureCRT) to connect to the corresponding COM port with these parameters:
 - **Baud Rate:** 115200
 - **Data Bits:** 8
 - **Stop Bits:** 1
 - **Parity:** None
 - **Flow Control:** None

4.3. Hardware Interface and Configuration Details

This section defines all key interfaces, connectors, and hardware configuration options on the EVB-A2.

4.3.1. Power and Reset System

- **Power Input (DJ1):** DC 12V/5A, center positive.
- **Power Switch:** Controls main board power.
- **Reset Circuit:** Supports multiple reset methods, selected via **Jumper Cap J80**:
- **Pins 1-2 shorted:** Button Reset + Reset IC (Default, more stable).
- **Pins 3-4 shorted:** Button Reset only.
- **System Reset Output:** The SoC's RST_OUT_N signal can be shorted via **Jumper Cap J88** to synchronize the reset of external circuits when the SoC resets.

The EVB-A2 defaults to Button + Reset IC mode. The jumper cap configuration and default SW1/SW3 settings are shown below. General users do not need to modify these:

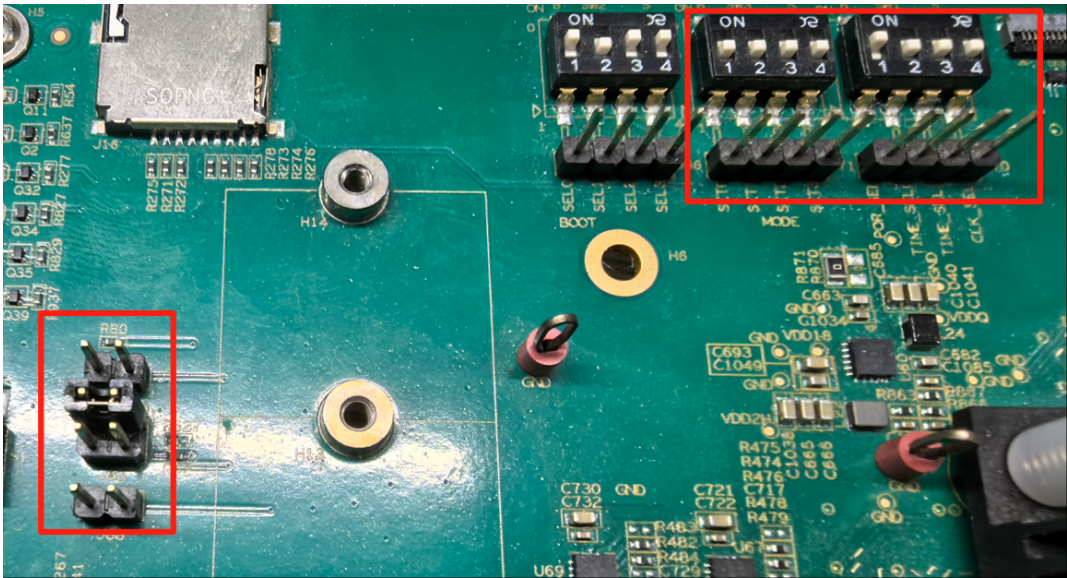


Figure 7: Default Reset Configuration

4.3.2. USB Interfaces

The EVB-A2 have 2 standard USB Type-A ports and 1 USB Type-C debug port, laid out as follows:

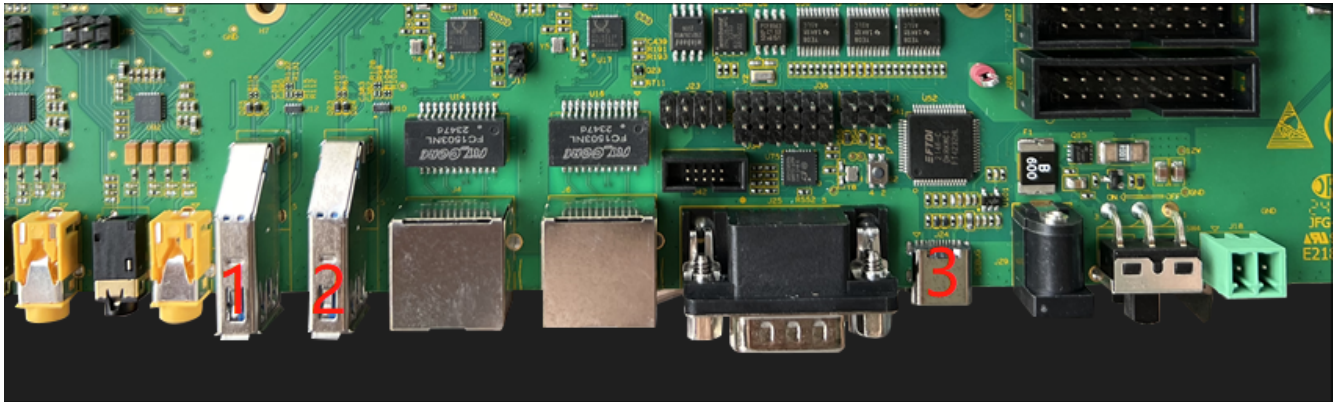


Figure 8: USB Interfaces

Table 2: SBC USB Interface Function Description

Port Location	Name	Function Description
Port 1	USB1	Connect USB 3.0 devices. <i>Note: In U-Boot, this port acts as a Host for USB drives.</i>
Port 2	USB0	Special function port: 1. In forced recovery mode, acts as a software download interface. 2. In U-Boot, can act as Device mode for Fastboot flashing.
Port 3	USB Type-C	Debug Serial Port: Connect to PC for serial debugging and login.

4.3.3. Core Expansion Interfaces

4.3.3.1. PCIe Gen3 x4 Expansion Slot

- **Spec:** PCIe x16 physical slot, supporting Gen3 x4 electrical signals.
- **Operating Mode:** Supports RC (Root Complex) and EP (Endpoint) modes. Default is RC mode.
- **Mode Switching:**
 1. Modify power supply method. RC is the power supplier, EP is the power receiver; in EP mode, disconnect beads FB42, FB43, FB44. Component locations are shown below.

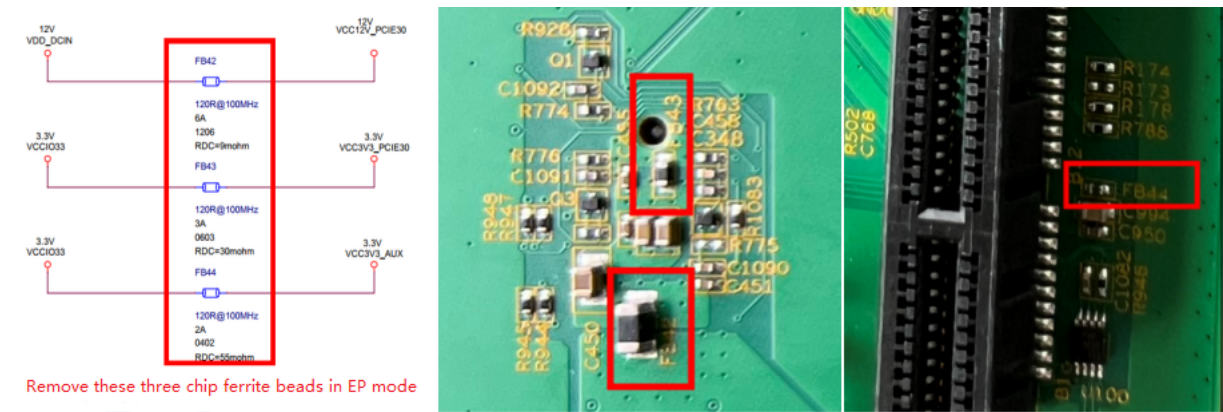


Figure 9: PCIe Power Beads and Location

2. PRSNT signal setting. PRSNT settings differ for RC and EP. The PRSNT detection principle and resistor settings are shown below.

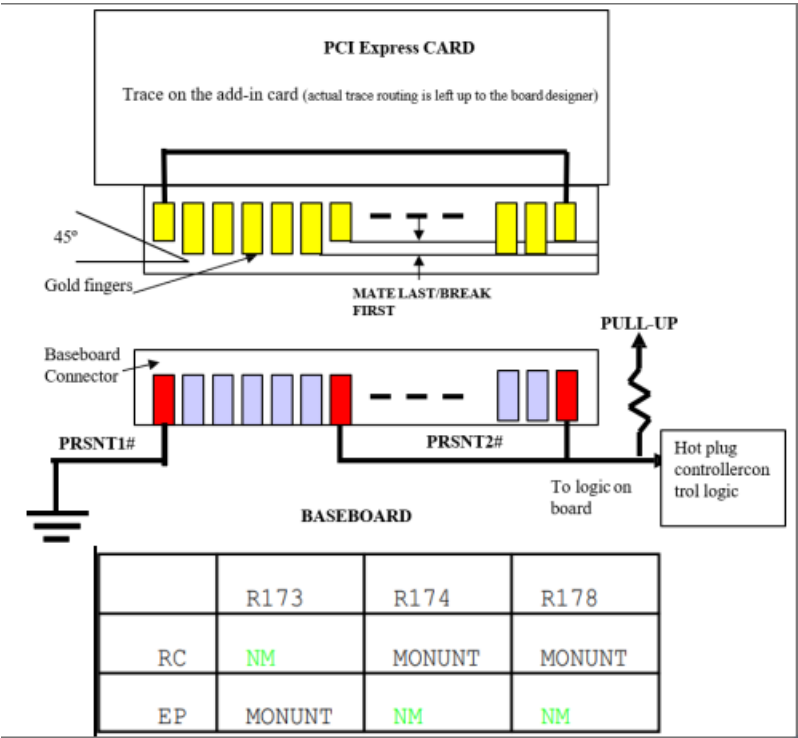


Figure 10: PRSNT Signal Settings

3. 100MHz reference clock source differs. Generally, RC provides the 100MHz clock to EP, so the 100M clock direction must be changed for different modes. EVB-A2 defaults to RC mode, with resistors

R523, R524, R525, R526 soldered. For EP mode, move these 4 zero-ohm resistors to positions R519, R520, R521, R522.

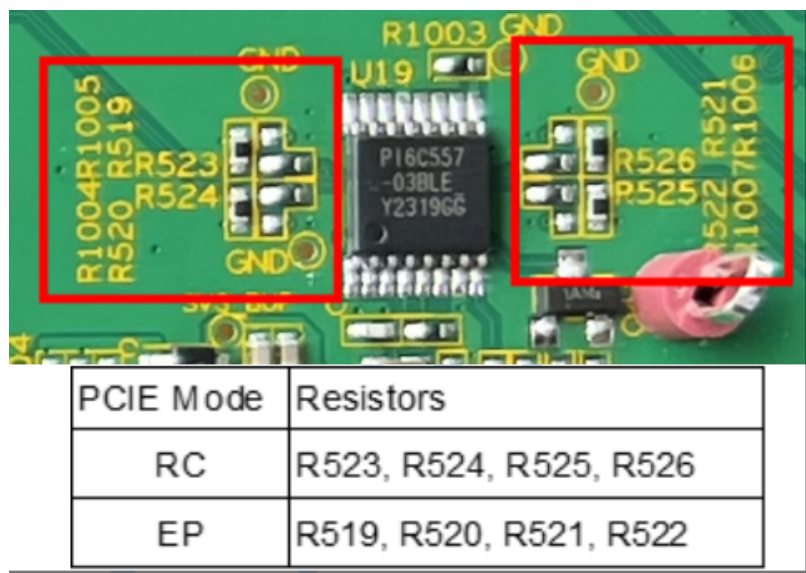


Figure 11: PCIE Clock Source Config Resistor Location

4.3.3.2. SATA Storage Interface

- **Interface:** Defaults to on-board M.2 SATA interface (J77), compatible with 2242/2280 sizes.

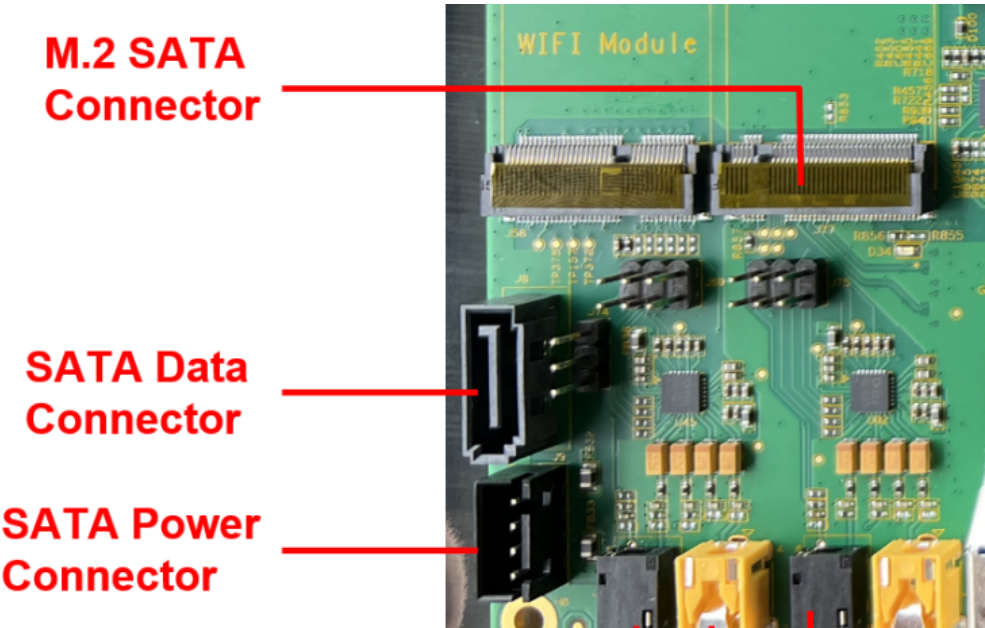


Figure 12: SATA Interface Location

- **Mode Switching:**
 1. To use the external SATA interface (J8), move resistors R834-R837 to positions R838-R841. These are all zero-ohm resistors.

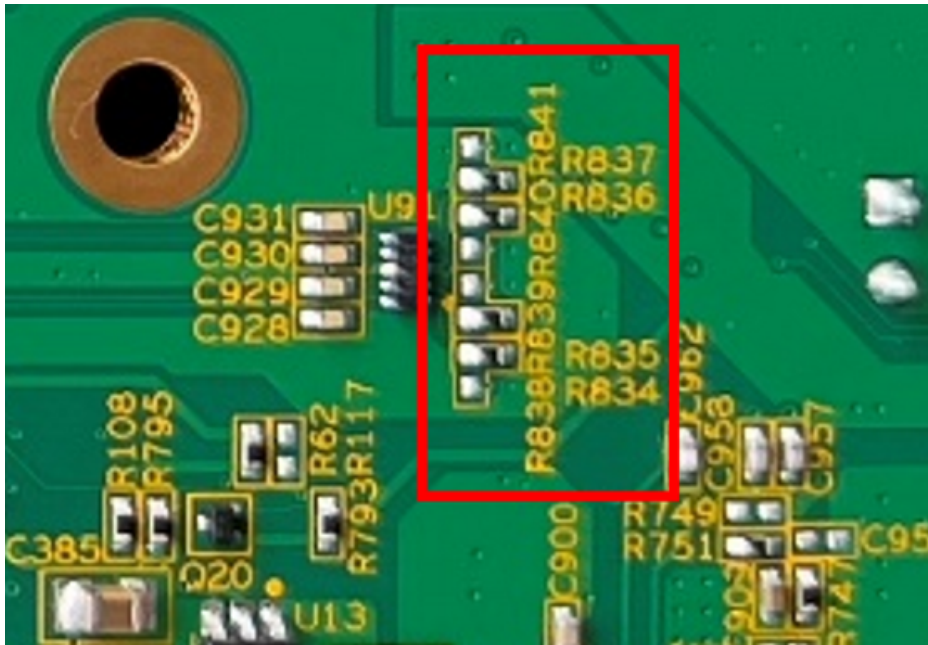


Figure 13: SATA Resistor Location

2. In addition to modifying resistors, a SATA adapter cable is needed to connect an external hard drive.

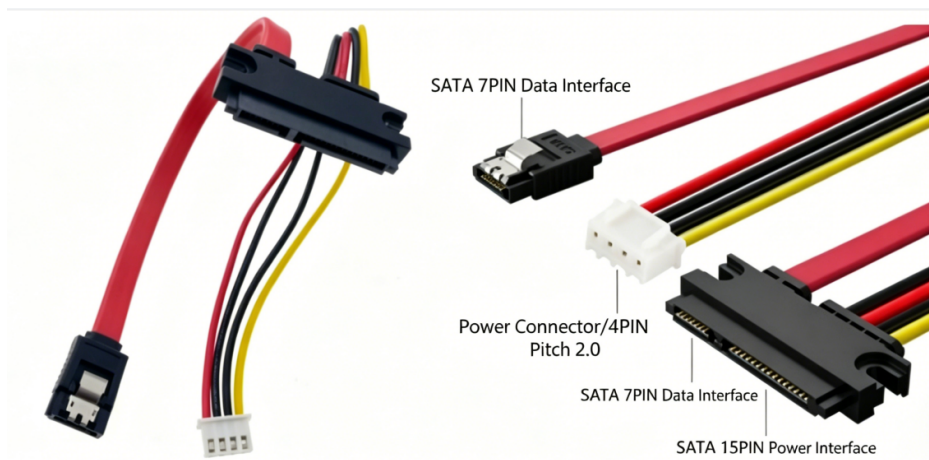


Figure 14: SATA adapter cable

4.3.4. Camera and Display Interfaces

4.3.4.1. MIPI CSI Camera Interface (J13, J14, J15)

- **Quantity:** 3 independent 4-lane MIPI CSI input interfaces.



Figure 15: CSI Port Location Diagram

- **Power Supply:** Camera module supply voltage (3.3V/5V) can be selected via **Jumper Cap J31**.

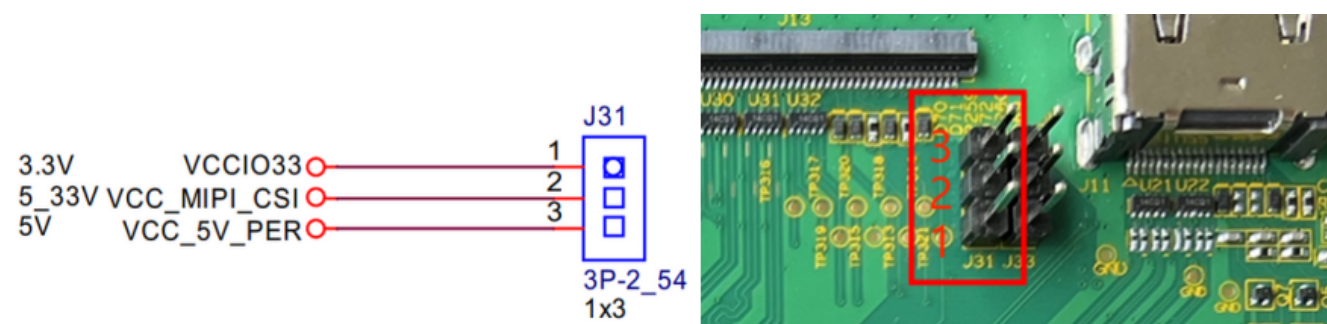


Figure 16: CSI Sensor Power Selection

Matching modules generally use 5V power. Jumper cap as shown:

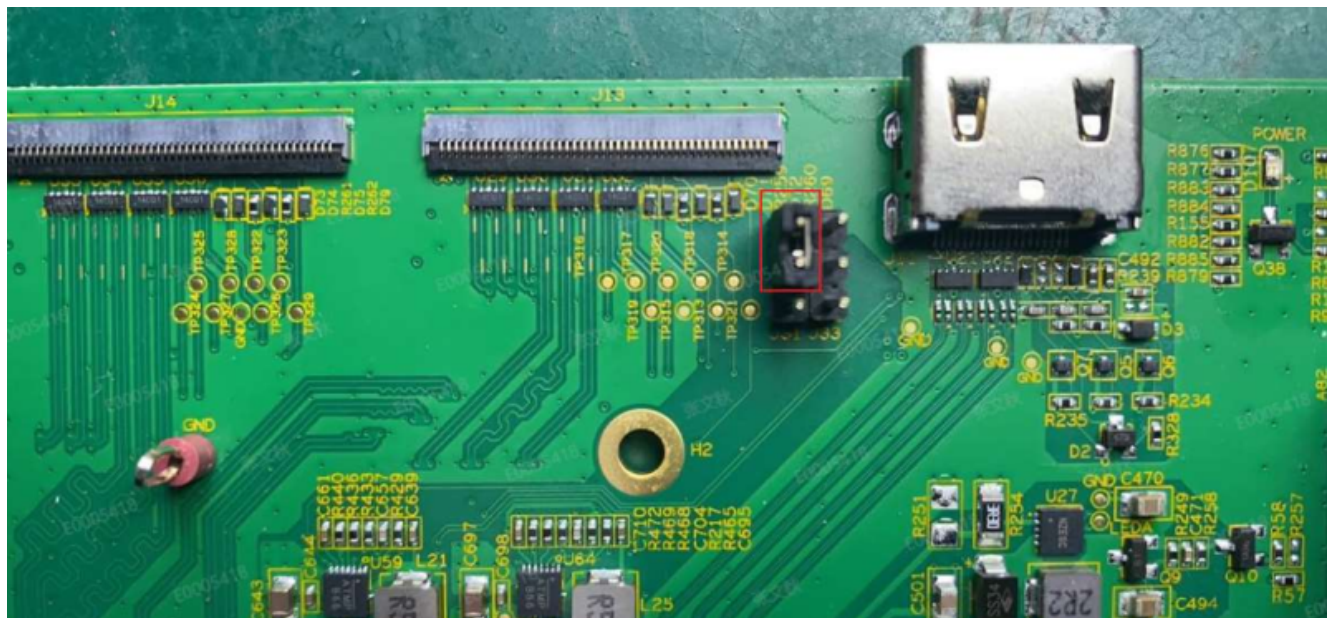


Figure 17: CSI Sensor 5V Power Selection

- **Connection:** Use FPC ribbon cables to connect camera modules, paying attention to the gold finger direction.

4.3.4.2. MIPI DSI Display Interface

- **Spec:** 4-lane MIPI DSI TX output (J12) and touch screen interface (J30).
- **Connection:** Used to connect MIPI displays.

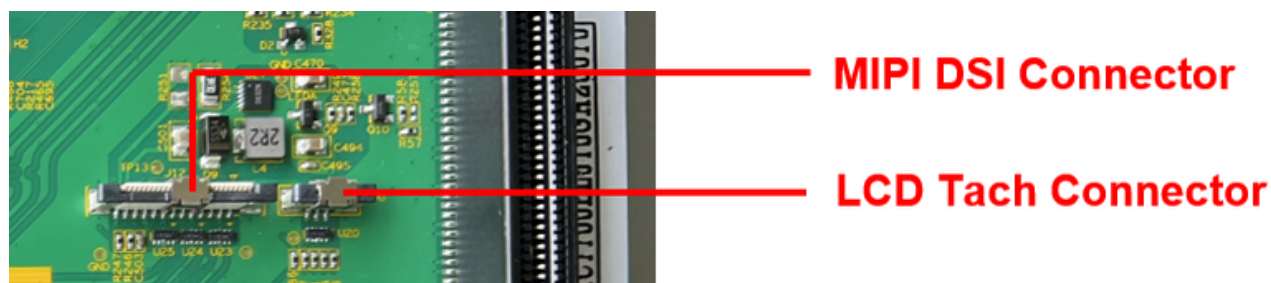


Figure 18: DSI Interface Location Diagram

4.3.5. Audio Interfaces

The EVB-A2 supports 2 channels of I2S. Each I2S channel has 1 Headphone and 1 Headset jack, and the I2S circuits are equipped with test headers, positioned as shown.

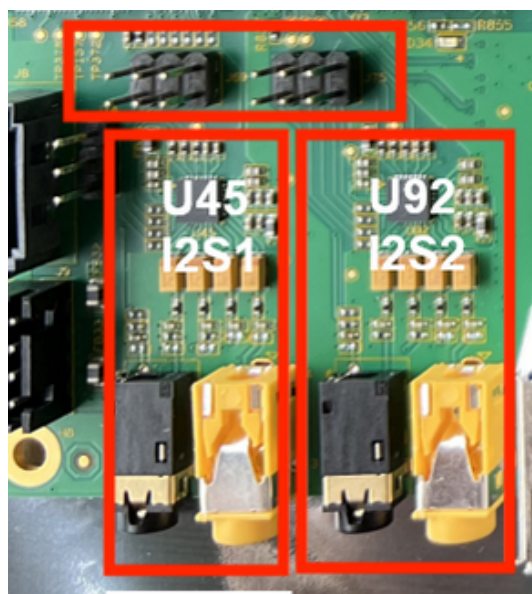


Figure 19: Audio Interfaces

4.3.6. Fan Interfaces

- **Interfaces:** Provides one 4-pin 12V fan header (J72) and one 3-pin 5V fan header (J73). Defaults to using the 5V fan header (J73).
- **Voltage Configuration:** Fan interface voltage can be selected via resistors R69 and R70.

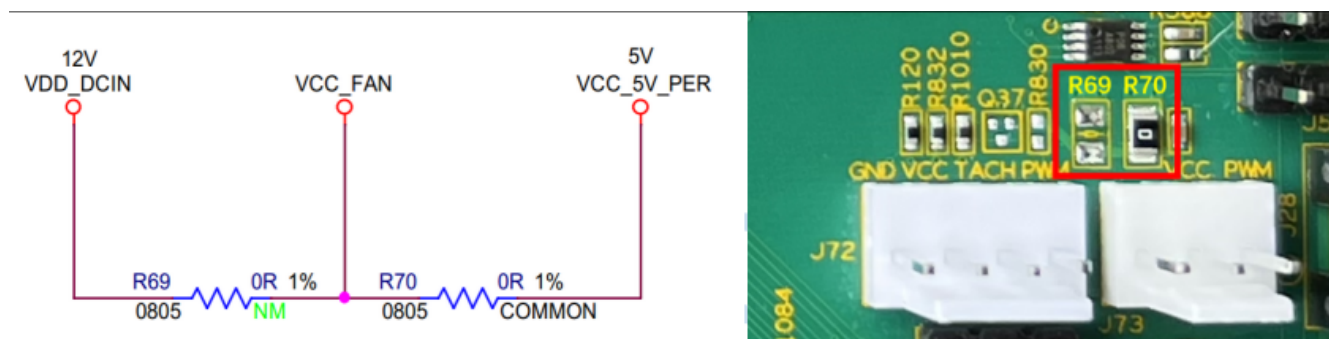


Figure 20: Fan Voltage Configuration

4.3.7. Debugging and Test Interfaces

- **Test Points:** The board has multiple test points (e.g., J42 for UART0, J24 for MCU JTAG/UART), mainly for advanced debugging and R&D. General users generally do not need to operate these.

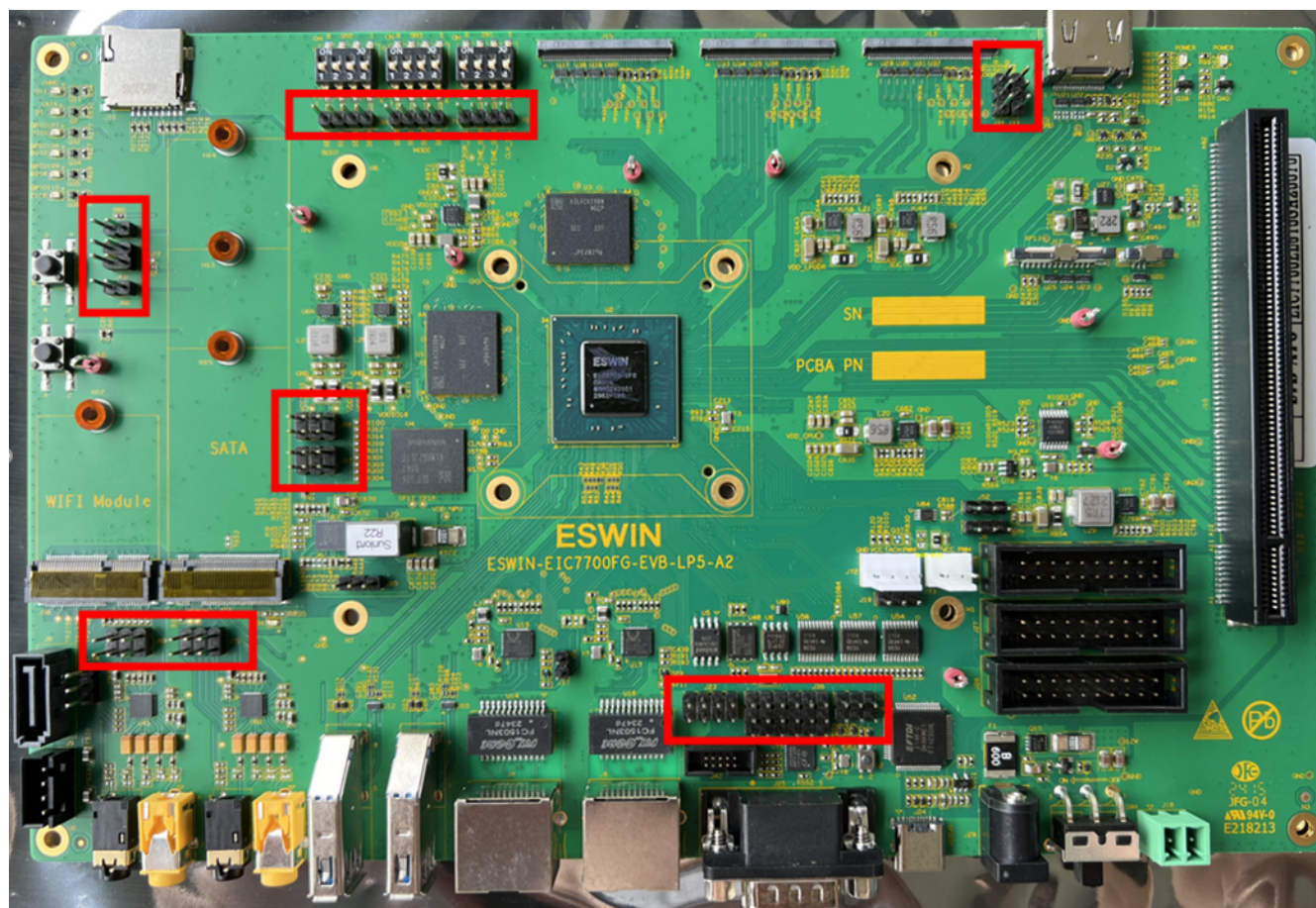


Figure 21: Test Point Location

5. Windows Driver Installation and Debugging Guide

Before using the EVB-A2 development board for serial debugging, console login, or Fastboot flashing under the Windows operating system, you must install the corresponding USB drivers. Linux users do not need to install additional drivers.

This chapter guides you through the installation of the following core drivers:

1. **FT4232H Driver:** Used to enable the Serial Port (COM) for the Type-C Debug Interface.
2. **Fastboot Driver:** Used for fast flashing of system images via USB.

5.1. FT4232H Driver Installation (Serial Port & JTAG)

The USB Type-C debug interface on the rear panel of the EVB-A2 converts a single USB connection into four independent virtual devices via the on-board FT4232H chip: two JTAGs and two UARTs, corresponding to the EIC7700 SoC and MCU respectively.

5.1.1. Obtain the Driver Installation Tool

1. Click to download the driver installation tool: [UsbDriverTool-2.1.exe](#).
2. Save it locally (e.g., to `D:\`), double-click to run, and extract it to the default UsbDriverTool directory.

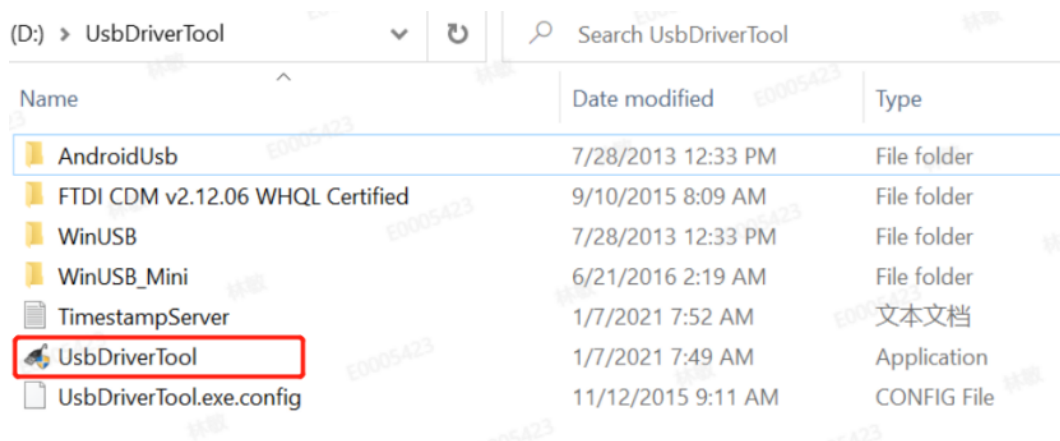


Figure 22: UsbDriverTool

5.1.2. Install the Serial Port (UART) Driver

1. Connect the rear panel Type-C port of the EVB-A2 to your Windows PC using a USB Type-C cable from the **Accessories**.
2. Open the extracted UsbDriverTool directory from the previous step and double-click UsbDriverTool.exe.
3. The program will automatically identify four `DeviceID` undriven devices with the ID `6011` (`00-03`). The EVB-A2 only uses the device with `Interface` `2` .
4. Right-click on the selected device and choose `Install FTDI CDM` from the pop-up menu.

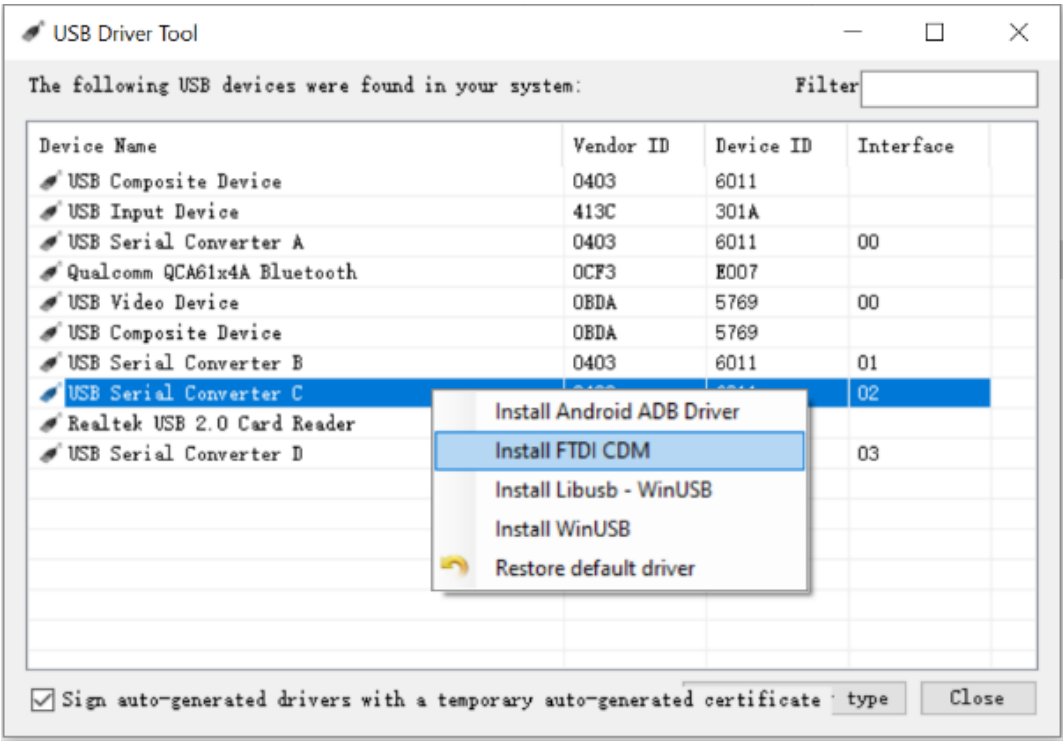


Figure 23: Install UART Driver

5. Verification of Installation:

Open the Windows Device Manager and expand the Ports (COM & LPT) list. You may see four new serial ports appear, such as COM3 - COM6.

- Typically, the third port in the list (e.g., COM5) corresponds to the UART of the EIC7700 SoC.



Figure 24: Serial Port Device

6. If the device manager displays USB Serial Port as shown below, right-click and select Update driver, then choose Browse my computer for driver software and select the FTDI CDM v2.12.06 WHQL Certified directory.

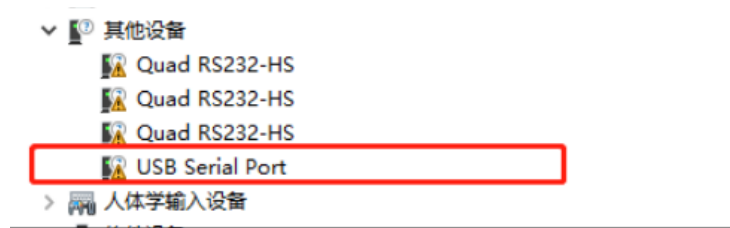


Figure 25: USB Serial Port

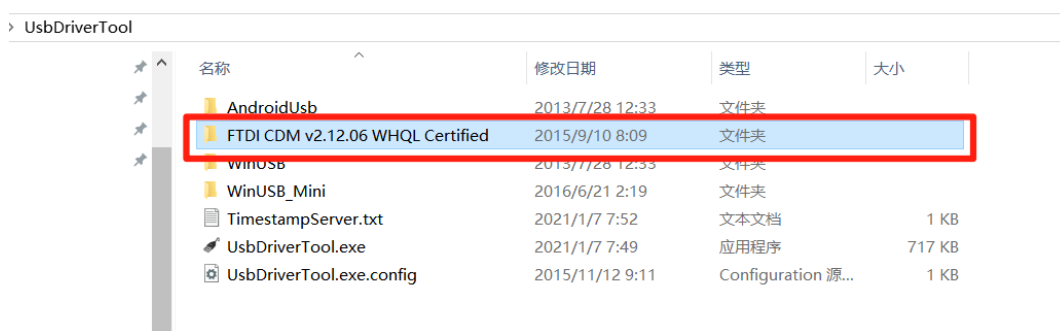


Figure 26: Driver Directory

Tip

The specific COM port number (e.g., COM5) varies depending on the computer; please refer to the actual display in the Device Manager.

Perform Serial Port Debugging:

After the driver is installed successfully, you can access the board using a serial port tool:

1. Open your preferred serial terminal software (e.g., Putty).
2. Select the corresponding COM port (for the main system, usually the third port, e.g., COM5).
3. Set the baud rate to 115200.
4. After connecting, power on the development board. You will see the boot information in the terminal and can perform command-line interaction.

5.2. Fastboot Driver Installation (System Flashing)

Fastboot is used to quickly flash the boot and root partition images of the Debian system via USB.

Note

In addition to Fastboot flashing, ESWIN also provides a USB drive flashing method. If you are not using Fastboot, please skip this section.

5.2.1. Prerequisites**1. Obtain Tools:**

- **Fastboot CLI Tools:** Download the `platform-tools-latest-windows.zip` archive from Google's repository. Extract it and add the `platform-tools` directory to the system's `PATH` environment variable.

- **Windows USB Driver:** Download the [Android USB Driver for Windows](#) (Version 1.9.0.0, 35.5MB) from the Samsung Developer website for later use.

2. Physical Connection:

- Use a USB Type - A male - to - male cable to connect the EVB-A2 **USB0** port to the PC.
- Ensure the Type-C debug port is also connected to the PC to access the U-Boot command line.

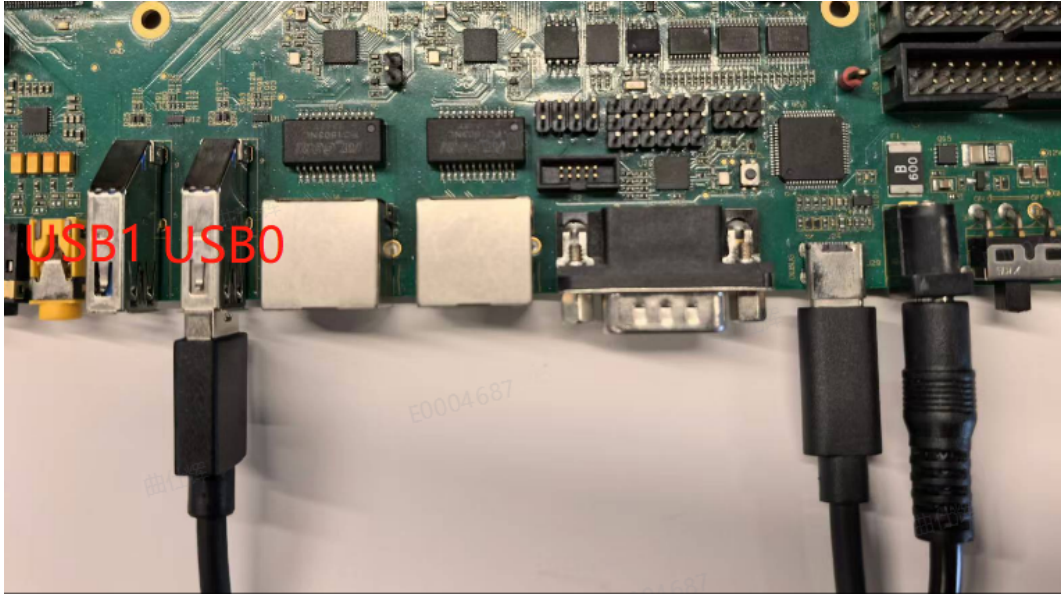


Figure 27: USB0

Note

The EVB-A2 has two USB Type-A ports:

- **USB0 (near the Ethernet port):** Configured as USB Device (Slave) mode in U-Boot, dedicated to Fastboot communication.
- **USB1 (away from the Ethernet port):** USB Host mode, used to connect peripherals like USB drives.

Please ensure you are connected to the **USB0** port for Fastboot operations.

5.2.2. Enter Fastboot Mode and Install Driver

1. Power on the development board via the serial terminal (connected to SoC UART, e.g., COM5). Press **Enter** during the U-Boot boot countdown to interrupt the boot process and enter the U-Boot command line.
2. In the U-Boot command line, enter the following command to set USB0 as the Fastboot device mode. The command line will return a message similar to `generic_phy_get_bulk: no phys property`, indicating it has entered the waiting state.

```
=> fastboot usb 0
generic_phy_get_bulk : no phys property
```

3. At this point, in the PC's "Device Manager", an unknown device named **USB download gadget** will appear under "Other devices".

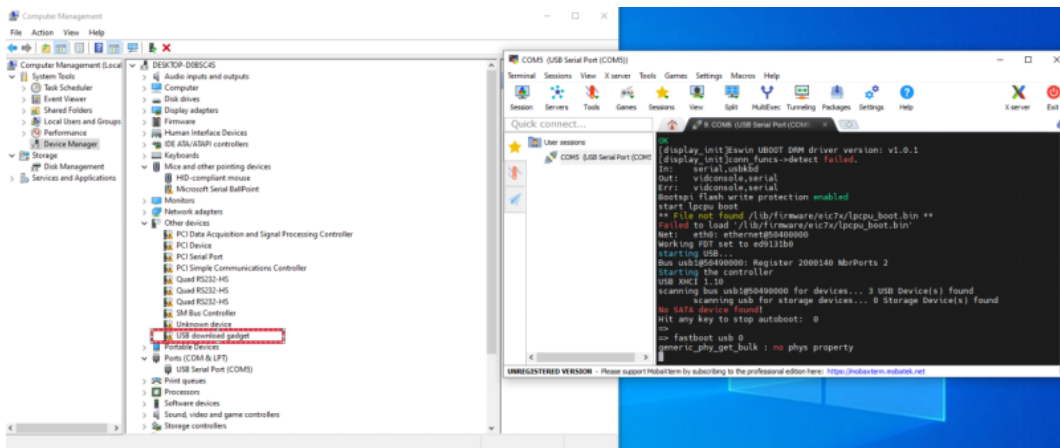


Figure 28: Fastboot Device Port

4. Right-click the **USB download gadget** device and select **Update driver**.

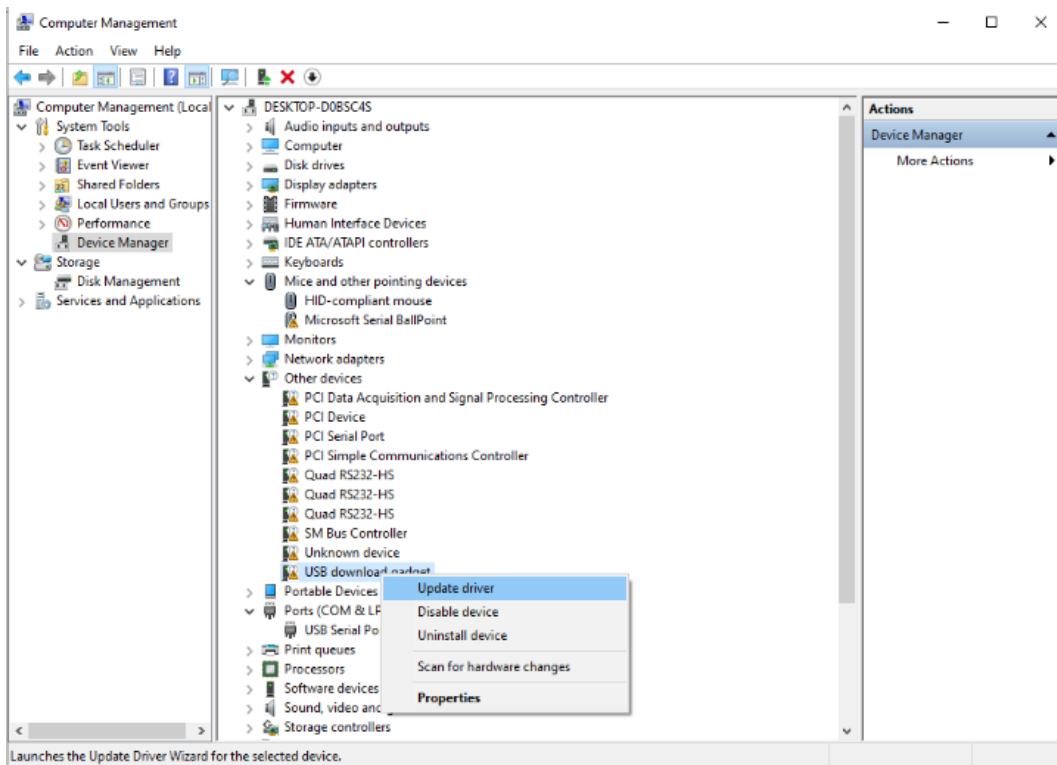


Figure 29: Fastboot Driver Update

5. In the pop-up window, select **Browse my computer for driver software**.

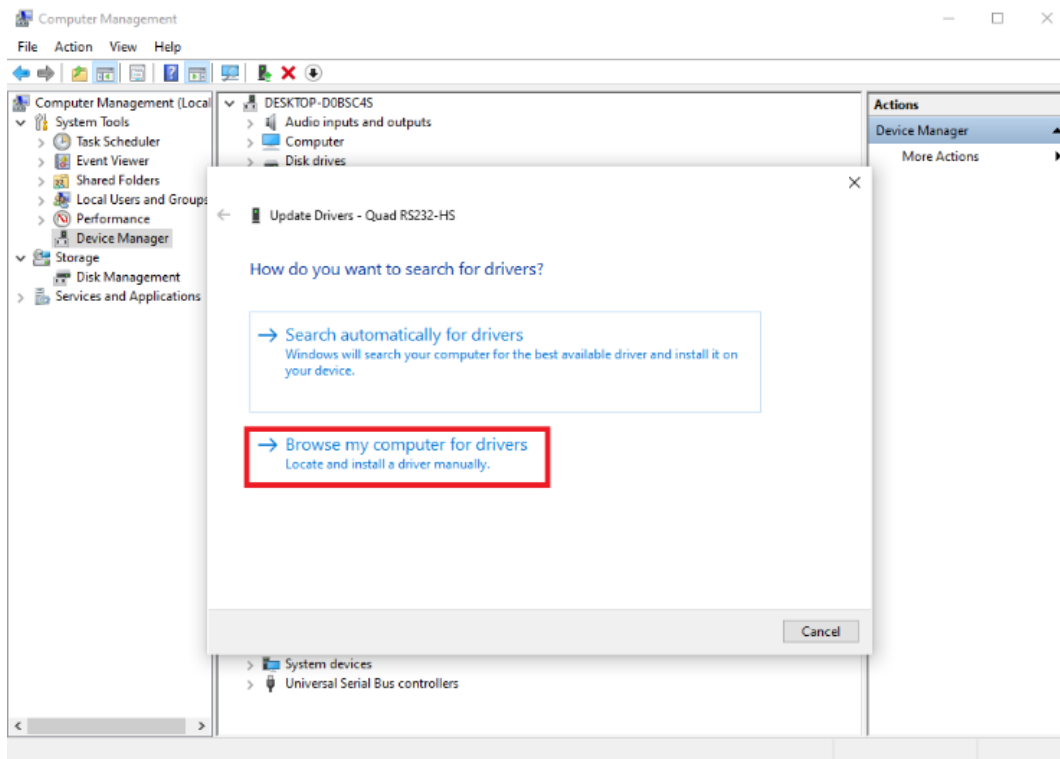


Figure 30: Fastboot Driver Browse

6. Select **Let me pick from a list of available drivers on my computer**.

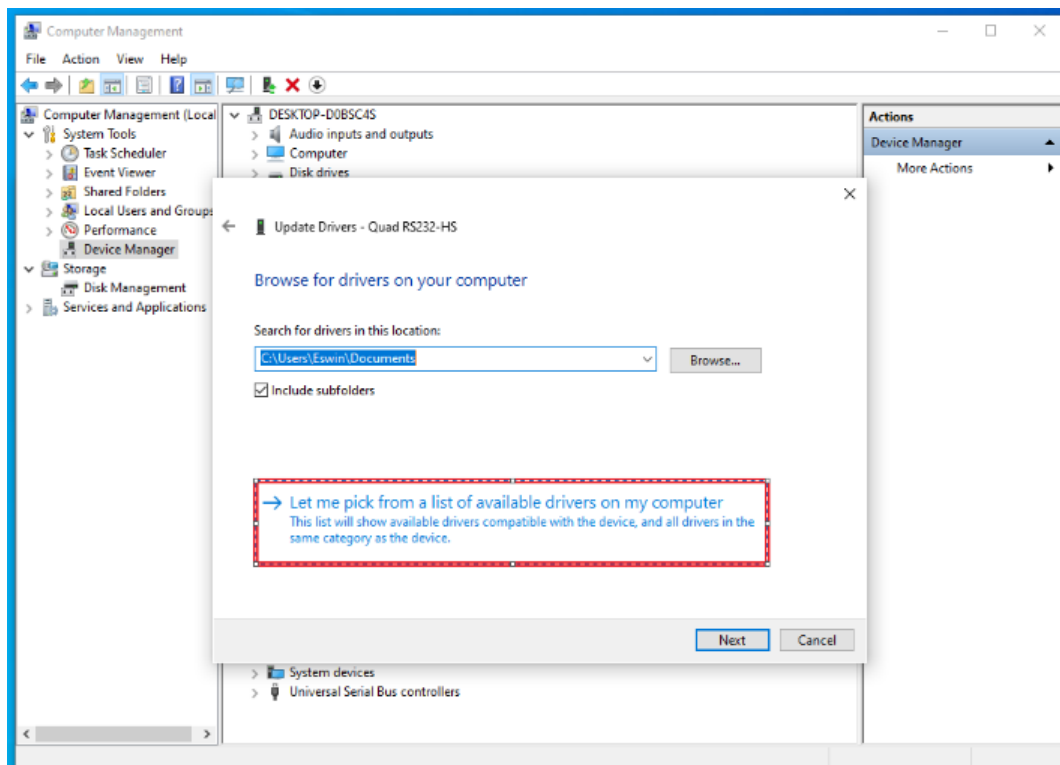


Figure 31: Fastboot Driver Select File

7. In the device type list, select **ADB Interface** and click Next.

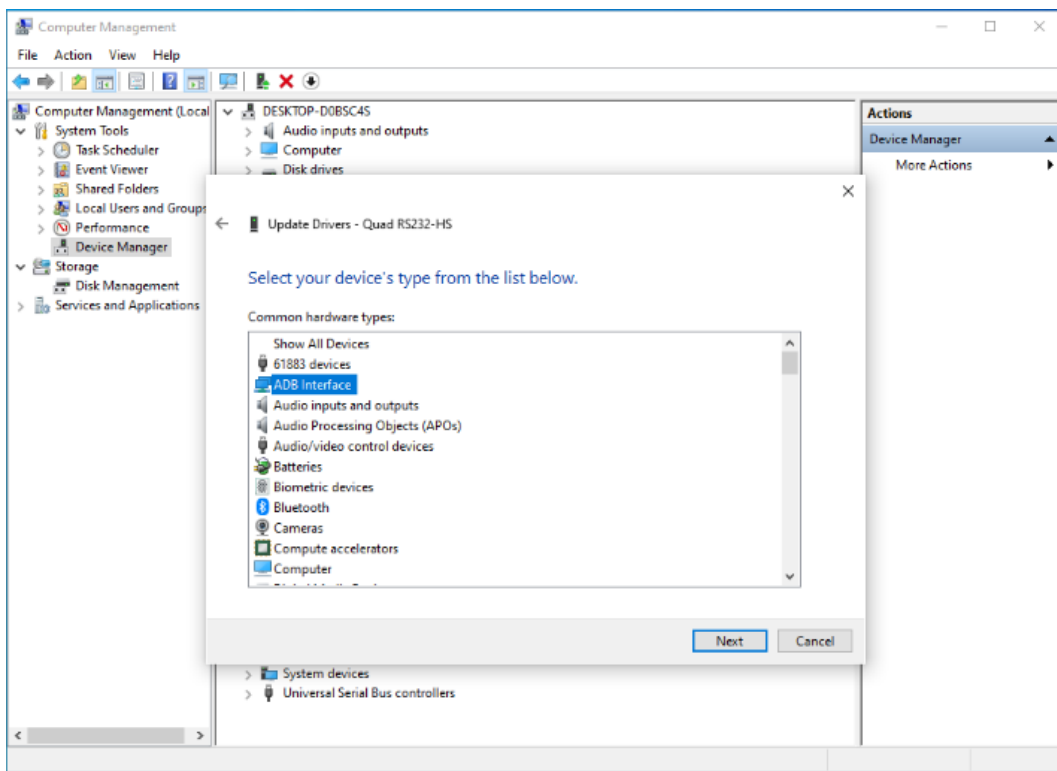


Figure 32: Fastboot Driver Select Device Type

8. Select the first driver and click Next to complete the installation.

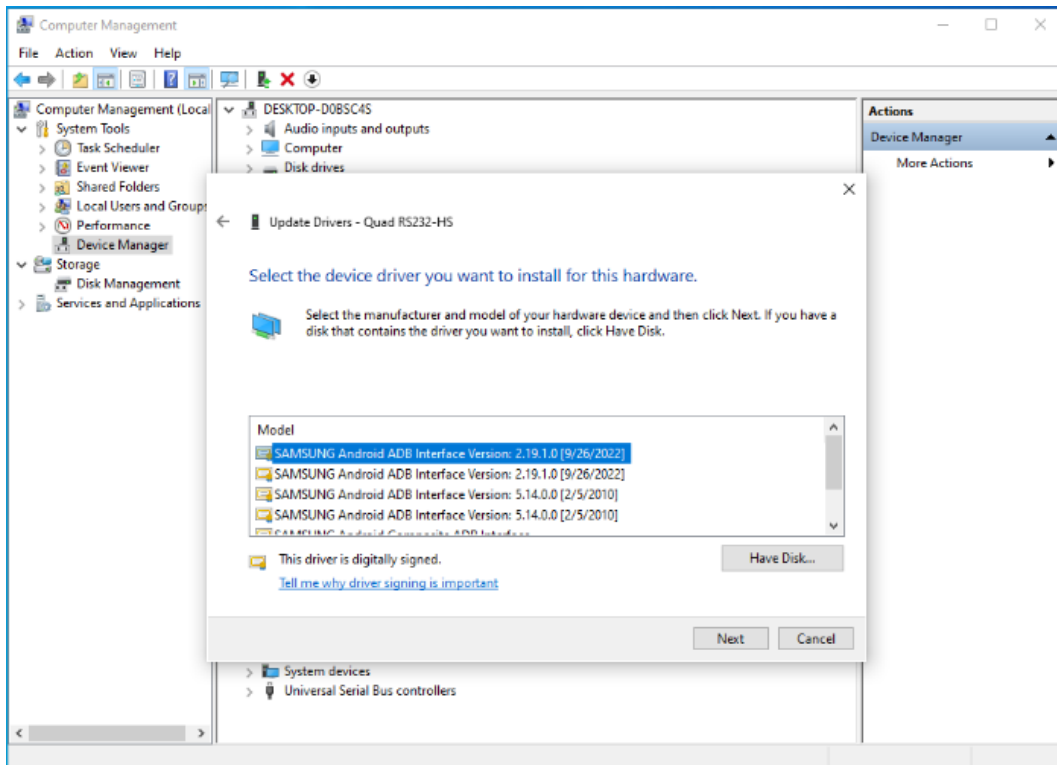


Figure 33: Fastboot Driver Select Driver

9. If the system pops up a warning Windows cannot verify this driver publisher , select Yes .

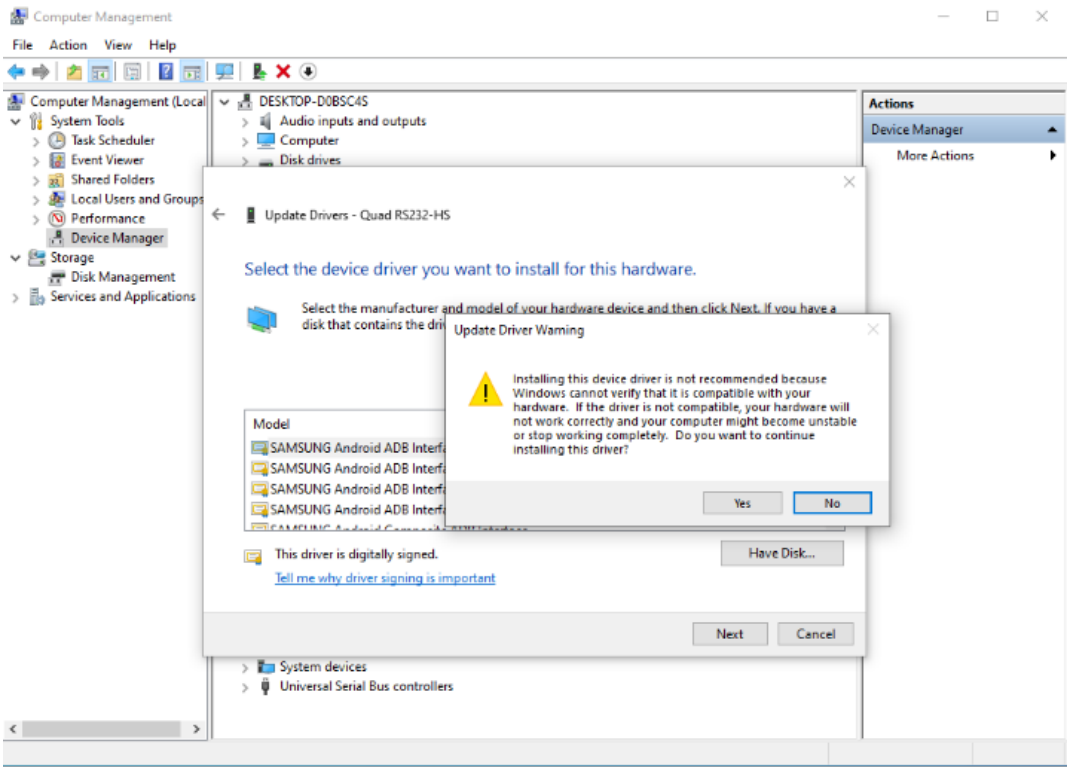


Figure 34: Fastboot Driver Confirmation

5.2.3. Verify Fastboot Connection

After the driver is successfully installed, a new USB device should appear under ADB Interface in the Device Manager .

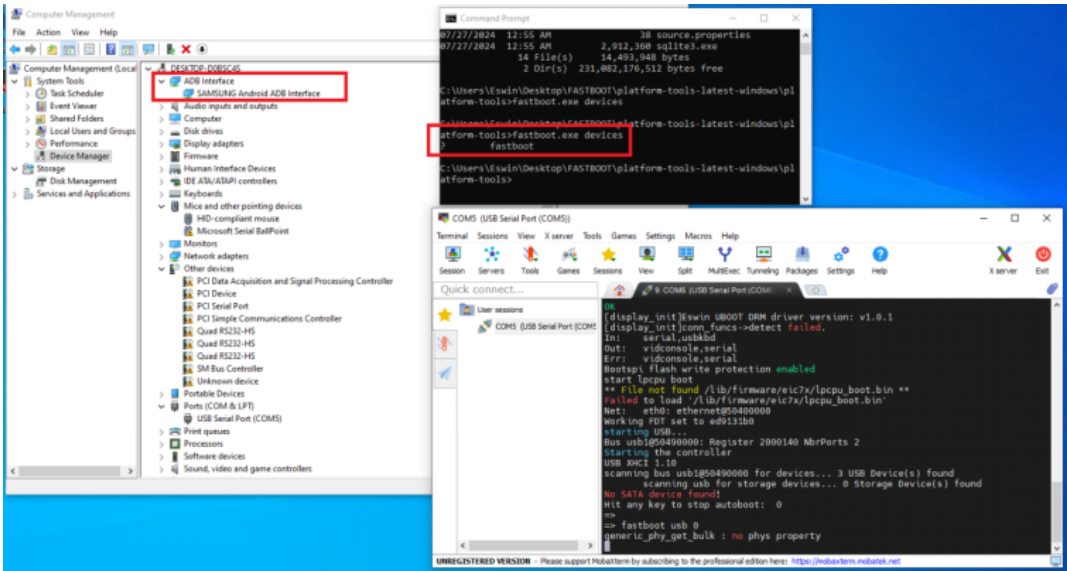


Figure 35: Fastboot Driver Refresh

- 1. Open the Command Prompt (CMD) or PowerShell on the PC.

2. Enter the following command to check if the device is recognized:

```
fastboot devices
```

3. If the connection is successful, the command line will return the device serial number with the status `fastboot`, for example:

```
xxxxxxx fastboot
```

The Fastboot driver installation is now complete. You can start using Fastboot commands to flash the system image.

6. System Image Flashing Guide

This chapter guides you on how to flash or update the system software for the EVB-A2. The EVB-A2 runs the Debian operating system and adopts a **Hybrid Boot Architecture**: U-Boot (Bootloader) is stored in the on-board SPI Flash, while the complete boot and root system are stored in the eMMC.

Therefore, the flashing operation is divided into two parts:

1. **Bootloader Flashing/Recovery**: Targets the on-board SPI Flash, used to boot the system.
2. **Debian System Flashing**: Targets the eMMC, containing the operating system itself.

Before starting, please ensure you have:

1. Completed all preparations in the [Hardware Introduction and Usage Guide](#), especially the installation of the heatsink.
2. Installed the necessary serial port drivers according to your development environment (Windows/Linux).
3. Prepared the following cables:
 - **USB Type-A Male-to-Male Data Cable** (for Forced Recovery Mode)
 - **USB Type-C Data Cable** (for Serial Debugging)

The EVB-A2 supports updating the Bootloader via USB Forced Recovery Mode and flashing the Debian system via U-Boot commands.

6.1. Obtaining the System Image

ESWIN provides pre-compiled Debian system images for the EVB-A2 officially.

1. Download the image: Download the `EVB-xxxx-debian.tar.zst` compressed package from the [official website](#) or designated resources to your local machine.
2. Extract the image: Use a tool that supports the zstd compression format to extract (the following takes Linux command line as an example):

```
tar --zstd -xvf EVB-xxxx-debian.tar.zst
```

3. Obtain Core Files: After extraction, you will get an `EVB/` directory containing the following 3 core files:

File Name	Functional Description	Key Role & Flashing Target
bootloader_EIC7700FG-EVB-LP5-A2.bin	Bootloader (U-Boot) Main Image	The boot program that runs first after system power-on. Used to upgrade an existing available Bootloader. Flashing Target: SPI Flash.
recovery_bootloader_EIC7700FG-EVB-LP5-A2.bin	Bootloader Recovery Image	Used for system recovery when the Bootloader in SPI Flash is corrupted and cannot boot normally.
boot-EIC7700FG-EVB-LP5-A2-YYYYMMDD-HHMM.ext4	Linux Kernel Partition Image	Contains the Linux kernel, device tree (dtb), and initial ramdisk (initrd), flashed to eMMC or SD card.
root-EIC7700FG-EVB-LP5-A2-YYYYMMDD-HHMM.ext4	Linux Root Filesystem Image	Contains the complete OS files, tools, and user-space programs, flashed to eMMC or SD card.

Tip

If you need to customize the system based on specific requirements (such as kernel configuration, drivers, packages), please refer to the separate [BSP Quick Start Guide](#) for system compilation.

Danger**Pre-upgrade Must-Read: Bootloader Pre-upgrade Guide**

Background: The new version EIC7X-26.06 introduces a Bootloader A/B backup mechanism to prevent bricking during upgrades. Due to changes in the partition structure and the old Bootloader's lack of support for the new A/B switching logic, old versions (20251230 and earlier) cannot be directly overwritten using the `es_burn` command. You must manually upgrade the Bootloader first.

Operation Paths (choose one):

1. Option 1 (Recommended): Refer to [Scenario 2: Bootloader fails to boot\(USB Mode Recovery\)](#) and complete the upgrade through the repair process.
2. Option 2 (Manual Tool): Use the dedicated bootloader-tool for command-line flashing.
 - Extract the `bootloader_EIC7700FG-EVB-LP5-A2.bin` file from the image package `EVB-26.06-debian.tar.zst`, and download the `bootloader-tool` file from the official website.
 - After booting the old system, transfer these two files to the device system directory (e.g., `/home/eswin/`).
 - Open a terminal, navigate to the file directory (e.g., `cd /home/eswin/`), and execute the following commands in order:

```
sudo ./bootloader-tool getinfo 0
sudo ./bootloader-tool upgrade bootloader_EIC7700FG-EVB-LP5-A2.bin
```

- After flashing is complete, please restart the device and enter Uboot mode.
- Once device connection is confirmed normal, you can proceed to upgrade the main system following the standard burning process.

6.2. Bootloader Flashing Guide

The flashing of the Bootloader is divided into two scenarios based on its current status:

- **Scenario 1: Bootloader boots normally** - Upgrade the existing Bootloader via U-Boot command line from a USB drive.
- **Scenario 2: Bootloader fails to boot** - Force recovery of the Bootloader via “USB Boot Mode” .

Please select the corresponding chapter to operate based on your actual situation.

Warning

The Bootloader is the cornerstone of system startup. Incorrect flashing steps, image files, or operational interruptions may cause the development board to fail to boot. Please strictly follow this guide and ensure stable power supply during operation.

6.2.1. Scenario 1: Bootloader Boots Normally (USB Drive Upgrade)

This method is suitable for routine upgrades where the Bootloader is running and can enter the U-Boot command line.

6.2.1.1. Preparation

1. **Prepare USB Drive:** Find a USB drive and format it to ext4 or FAT32 file system. The ext4 format is recommended because FAT32 has a 4GB file size limit for single files when flashing larger rootfs images later. U-Boot supports reading both formats.
2. **Place Image Files:** Copy the `bootloader_EIC7700FG-EVB-LP5-A2.bin` file to be upgraded to the root directory of the USB drive.

6.2.1.2. Operation Steps

1. **Connect & Boot:** Insert the prepared USB drive into the development board's USB1 interface (away from the Ethernet port). Boot the development board, press the Enter key during the U-Boot boot countdown interface to enter the U-Boot command line.

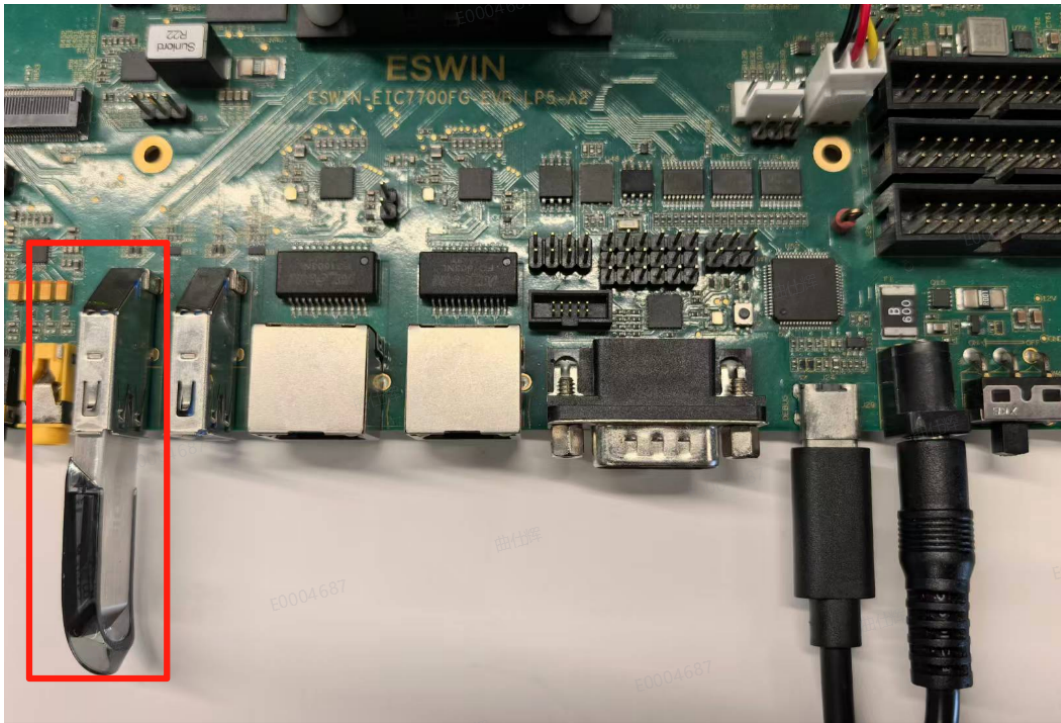


Figure 36: USB Drive Upgrade

2. **Identify USB Device:** In the U-Boot command line, enter the following command to initialize USB and scan storage devices.

```
=> usb reset
```

After the command is executed, please confirm that `1 Storage Device(s) found` appears in the output message. This indicates that the USB drive has been successfully recognized. If not found, please check if the USB drive is properly inserted, if the format is correct, and if it is plugged into the wrong USB port.

3. **List USB Drive Files:** Enter the following command to view the files in the root directory of the USB drive and confirm that `bootloader_EIC7700FG-EVB-LP5-A2.bin` exists.

```
=> ls usb 0
```

Tip

If the USB drive has multiple partitions and the image file is not on the first partition, you can use `ls usb 0:2` to view the second partition (and so on).

4. **Load Image to Memory:** Depending on the file system format of the USB drive, use the corresponding command to load the Bootloader image to the specified memory address (`0x90000000`).

- **ext4 format USB drive:**

```
=> ext4load usb 0 0x90000000 bootloader_EIC7700FG-EVB-LP5-A2.bin
```

- **FAT32 format USB drive:**

```
=> fatload usb 0 0x90000000 bootloader_EIC7700FG-EVB-LP5-A2.bin
```

After successful loading, the terminal will display the number of bytes read and the speed.

5. **Flash Image to SPI Flash:** Execute the following key command to write the image from memory to the on-board SPI Flash.

```
=> es_burn write 0x90000000 flash
```

This process will output detailed erase and write progress. Please be sure to wait until the final prompt `bootloader write OK` appears, indicating that the flashing was successful.

Tip

```
=> es_burn write 0x90000000 flash
Bootspi flash write protection disabled
SF: 4096 bytes @ 0x1000 Read: OK
SF: 4096 bytes @ 0x0 Read: OK
UPDATE BOOTLOADER FAILED (Invalid image file type)
Bootspi flash write protection enabled
exit not allowed from main input shell.
```

If the above error message appears, please erase the flash first and then write again. The operation steps are as follows:

- Erase command: `es_burn erase flash`
- Write command: `es_burn write 0x90000000 flash`

6. **Reboot System:** After the flashing is complete, enter the following command to reboot the development board, and the new Bootloader will take effect immediately.

```
=> reset
```

6.2.2. Scenario 2: Bootloader fails to boot(USB Mode Recovery)

When the Bootloader fails to start, you need to set the development board to USB Device Mode to recover it directly from the host (PC).

Warning

The method in this section is **only** for repairing a “bricked” device where the Bootloader is corrupted and no command line can be accessed. If the Bootloader can start normally, please prioritize the safer USB drive upgrade method described above.

6.2.2.1. Preparation

1. **Cables:** Use the USB Type-A data cable included in the accessories.
2. **Prepare Recovery Image:** Place the `recovery_bootloader_EIC7700FG-EVB-LP5-A2.bin` file in an easily accessible location on your PC.

6.2.2.2. Operation Steps

1. **Hardware Connection:** Use the dual-head USB Type-A male - to - male cable to connect the development board's USB0 interface (near the Ethernet port) to the PC.

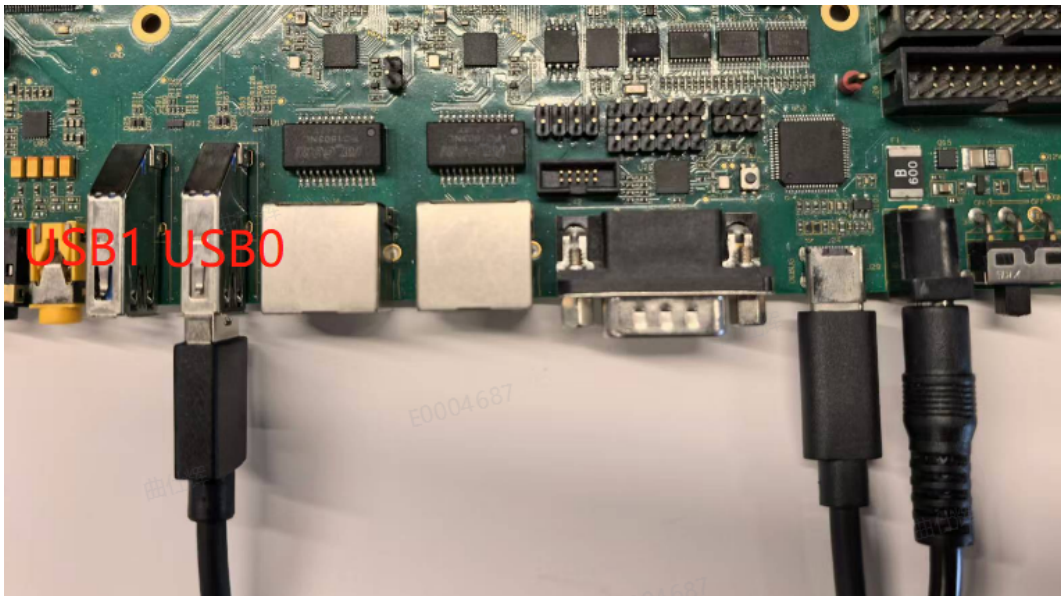


Figure 37: USB Mode Recovery

2. **Set USB Boot Mode:**
 - Set the on-board DIP switch SW2 to `1100` (i.e., USB Boot Mode). For specific setup instructions, please refer to [Boot Mode Selection](#).
 - Connect the power supply and turn on the `Main Switch`.
3. **Boot Development Board:** After the development board boots, the PC will recognize it as a removable disk named `ESWIN-2030` (USB drive).

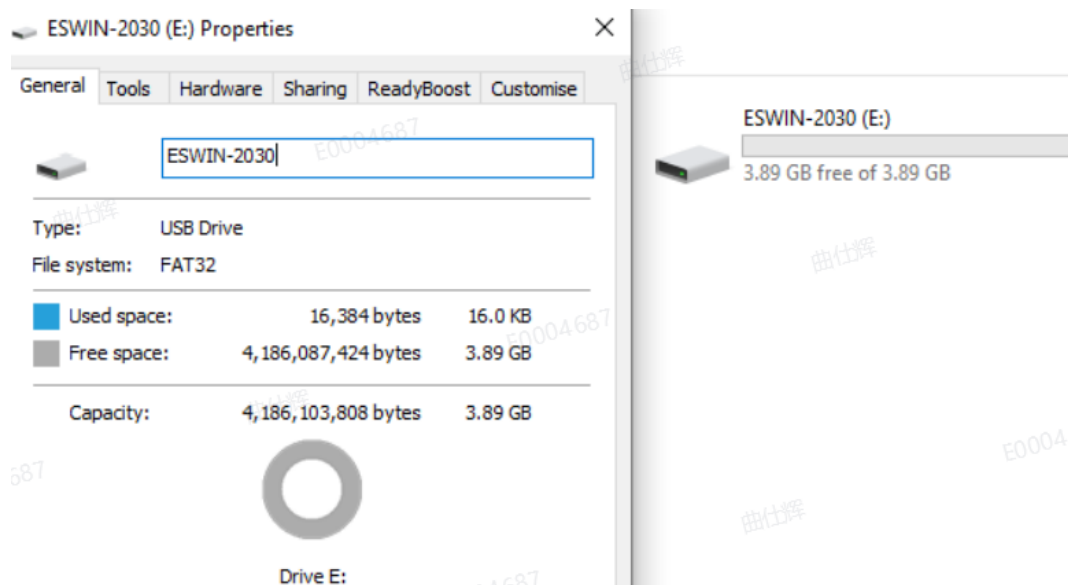


Figure 38: Simulated USB Drive Device

4. **Copy Recovery Image:** Copy the prepared file `recovery_bootloader_EIC7700FG-EVB-LP5-A2.bin` (or drag and drop it) into this disk.
5. **Wait for Automatic Recovery:** After the file copy is complete, the development board will automatically start the recovery process. You can observe the progress through two methods:
 - **HDMI Monitor:** The screen will display an `Upgrading...` progress bar, which changes to `Upgrade Finished` upon completion.
 - **Serial Terminal:** You can see detailed recovery logs via a serial tool (connected to SoC UART). The final indicator of successful recovery is the appearance of `bootloader write OK with logo` in the logs.
6. **Recovery Complete & Reboot:**
 - When the monitor indicates the upgrade is complete, or the serial logs show successful writing, the Bootloader has been recovered.
 - Turn off the development board power (turn off the `Main Switch` or unplug the power cord).
 - Reset the DIP switch `SW2` to `0100` (Boot from SPI Flash).
 - Power on again, and the development board will start normally using the recovered Bootloader.

6.2.2.3. Post-Recovery Operations

After the Bootloader is successfully recovered, the development board can boot normally. Next, you need to continue flashing the complete operating system for the development board.

6.3. Flashing Debian System Image

The Debian system image supports two mainstream flashing methods: USB Drive Flashing and Fastboot Flashing.

Tip

1. **USB Drive Flashing:** Only requires a formatted USB drive, suitable for all environments, especially recommended for beginners.
2. **Fastboot Flashing:** Requires driver installation and network configuration on the PC, suitable for users familiar with Fastboot tools.

6.3.1. Method 1: USB Drive Flashing Guide

This method writes the system image directly from a USB drive to the on-board eMMC storage via the U-Boot command line.

6.3.1.1. Preparation

1. **Prepare USB Drive:** Prepare a USB drive with a capacity of at least 16GB and format it to the ext4 file system (Single file exceeds 4GB in size).
2. **Obtain Image Files:** Download and extract the Debian system package from [Obtaining System Image](#) to get the following two files:
 - `boot-EIC7700FG-EVB-LP5-A2-YYYYMMDD-HHMM.ext4` (Kernel Partition Image)
 - `root-EIC7700FG-EVB-LP5-A2-YYYYMMDD-HHMM.ext4` (Root Filesystem Image)
3. **Place Images:** Copy the two `.ext4` image files mentioned above to the root directory of the USB drive.

6.3.1.2. Operation Steps

Please strictly follow the steps below in order.

1. **Connect USB Drive & Boot:** Insert the prepared USB drive into the development board's USB1 interface (away from the Ethernet port). Boot the development board, press the Enter key during the U-Boot boot countdown interface to enter the U-Boot command line.
2. **Identify USB Device:** In the U-Boot command line, enter the following command to scan USB storage devices:

```
=> usb reset
```

After the command is executed, please confirm that `1 Storage Device(s) found` appears in the output message. This indicates that the USB drive has been successfully recognized. If not found, please check if the USB drive is properly inserted, if the format is correct, and if it is plugged into the wrong USB port.

3. **Verify Image Files:** Enter the following command to list the files in the root directory of the USB drive and confirm that both system image files exist:

```
=> ls usb 0
```

Note

If the USB drive has multiple partitions and the image file is not on the first partition, you can use `ls usb 0:2` to view the second partition (and so on).

4. **Check Target Storage Partition:** Before flashing, you need to confirm that the partition table of the on-board eMMC (`mmc 0`) is correct. Execute the following command to view the partition information:

```
=> mmc part

Partition Map for MMC device 0 -- Partition Type: EFI

Part  Start LBA    End LBA    Name
Attributes
Type GUID
Partition GUID
1 0x00000800 0x001007ff "boot"
  attrs: 0x0000000000000000
  type: c12a7328-f81f-11d2-ba4b-00a0c93ec93b
      (system)
  guid: 44b7cb94-f58c-4ba6-bfa4-7d2dce09a3a5
2 0x00100800 0x009007ff "swap"
  attrs: 0x0000000000000000
  type: 0657fd6d-a4ab-43c4-84e5-0933c84b4f4f
      (swap)
  guid: 5ebcaaf0-e098-43b9-beef-1f8deedd135e
3 0x00900800 0x045007ff "root"
  attrs: 0x0000000000000000
  type: 0fc63daf-8483-4772-8e79-3d69d8477de4
      (linux)
  guid: b0f77ad6-36cd-4a99-a8c0-31d73649aa08
```

You need to specifically confirm whether the output includes the `boot` partition numbered `1` and the `root` partition numbered `3`. Correct partitions are a prerequisite for successful flashing.

Warning

- **If the partition information is correct:** Please skip directly to Step 5.
- **If partitions are missing or incorrect:** You must perform the partitioning operation first, otherwise flashing will fail or cause the system to be unbootable. Enter the following command to recreate the standard GPT partition table:

```
=> run gpt_partition
```

After seeing the `Writing GPT: success!` prompt, be sure to execute the `mmc part` command again to verify that the `boot(1)` and `root(3)` partitions have been created correctly.

5. **Flash Boot Partition Image:** Write the kernel image from the USB drive to the eMMC's `boot` partition (`mmc 0:1`). This process will take several minutes; please wait patiently until it is complete.

```
=> es_fs update usb 0 boot-EIC7700FG-EVB-LP5-A2-YYYYMMDD-HHMM.ext4 mmc 0:1
```

Success Indicator: `mmc has been successfully written in mmc 0:1` should be displayed at the end of the command output.

6. **Flash Root Partition Image:** Write the root filesystem image from the USB drive to the eMMC's `root` partition (`mmc 0:3`). This is the largest image and the most time-consuming; do not interrupt the process.

```
=> es_fs update usb 0 root-EIC7700FG-EVB-LP5-A2-YYYYMMDD-HHMM.ext4 mmc 0:3
```

Success Indicator: `mmc` has been successfully written in `mmc 0:3` should be displayed at the end of the command output.

7. **Completion and Reboot:** After both images have been successfully flashed, enter the following command to reboot the development board:

```
=> reset
```

The development board will boot from the newly flashed Debian system. The first boot involves system initialization and may take a longer time; please wait patiently until the login prompt appears.

6.3.1.3. Troubleshooting

1. **Error During Flashing:** Please re-check if the USB drive is connected securely, if the image files are intact, and if the eMMC partitions (Step 4) are correct.
2. **System Fails to Boot:** Please confirm that you have strictly followed Steps 5 and 6, flashing the images to the correct partitions (`mmc 0:1` and `mmc 0:3`). You can view the boot logs via the serial port to determine the failure stage.

6.3.2. Method 2: Fastboot Flashing Guide

This method uses a USB connection and the Fastboot tool to quickly flash images from a PC host.

6.3.2.1. Prerequisites

1. **Fastboot drivers must be installed on the PC** (see [Driver Installation Guide](#)).
2. **The development board and PC must be connected via two cables:**
 - **USB Type-C Cable:** For the serial console (to enter U-Boot).
 - **USB Type-A Cable:** For Fastboot data transfer (connect to the development board's USB0 interface).
3. **Obtain Image Files:** Same as USB flashing, prepare the `boot...ext4` and `root...ext4` image files on the PC.
4. **Prepare PC Environment:** Ensure the PC has the `fastboot` command-line tool installed and that its directory has been added to the system `PATH` environment variable.

6.3.2.2. Operation Steps

1. **Enter U-Boot and Check Partitions:**
 - Reboot the development board via the serial terminal (e.g., connecting to COM5) and press Enter during the U-Boot countdown to enter the command line.
 - Execute the `mmc part` command to check and confirm that the `boot(1)` and `root(3)` partitions exist and are correct. If repair is needed, execute the `run gpt_partition` command.
2. **Start Development Board Fastboot Mode:** In the U-Boot command line, enter the following command to set USB0 as the Fastboot device:

```
=> fastboot usb 0
```

The command line will return messages such as `generic_phy_get_bulk: no phys property`, indicating that the development board has entered the waiting connection state.

3. **Verify Fastboot Connection on PC:** Open the Command Prompt (CMD) or terminal on your PC and execute:

```
fastboot devices
```

If the connection is normal, it will return the device serial number with the status `fastboot`.

4. **Flash Boot Partition Image:** On the PC command line, navigate to the directory where the image files are located and execute:

```
fastboot flash boot boot-EIC7700FG-EVB-LP5-A2-YYYYMMDD-HHMM.ext4
```

Wait for the flashing to complete; the command line will display the transmission speed and an “OKAY” status.

5. **Flash Root Partition Image:** In the PC command line, continue to execute:

```
fastboot flash root root-EIC7700FG-EVB-LP5-A2-YYYYMMDD-HHMM.ext4
```

This is the largest image and is time-consuming; do not interrupt.

6. **Completion and Reboot:** After both images have been flashed, execute the reboot command in the PC command line:

```
fastboot reboot
```

The development board will automatically reboot and boot from the newly flashed system.

Tip

On Linux, please use `sudo fastboot` to execute commands.

6.3.2.3. Advanced Feature: Network (UDP) Fastboot

The Fastboot tool supports flashing via Ethernet UDP protocol, which may be faster and does not require a USB data cable.

- Connect the development board's Ethernet ETH0 (near the USB port) to the network cable, ensuring it is on the same local area network as the PC host and supports DHCP service.
- **In U-Boot:** Use the `fastboot udp 0` command to start the network Fastboot service and assign an IP address to the development board.
- **On PC:** Use `fastboot -s udp:<Development Board IP Address> flash` to specify the target device for flashing.
- For specific configuration steps, please contact ESWIN technical support (FAE).

7. Frequently Asked Questions (FAQ)

This chapter summarizes common issues and their solutions that you may encounter when using the EVB-A2 development board. Please refer to this section before use or when troubleshooting.

7.1. System Boot and U-Boot

Q1: How do I get familiar with U-Boot commands and functions?

A: In the U-Boot command line, you can get help at any time by:

- Typing `help` or `?` to see a list of all available commands.
- Typing `help <command_name>` to view detailed usage instructions for a specific command (e.g., `help mmc`).

Q2: What should I do if characters are truncated or lost when pasting long commands into the serial port?

A: This is a default limitation of the U-Boot serial input buffer. It can be resolved by modifying environment variables:

```
=> setenv stdout serial
=> saveenv
```

Warning

This setting will redirect all U-Boot output exclusively to the serial terminal, disabling output to the HDMI monitor. To restore HDMI output, execute `setenv stdout vidconsole,serial` followed by `saveenv`.

Q3: Does U-Boot support USB device hot-plugging?

A: Yes. After inserting a USB drive, you need to execute the `usb reset` command in the U-Boot command line to rescan and identify the device.

Q4: Does U-Boot support SD card device hot-plugging?

A: No. The SD card must be inserted before powering on the system; otherwise, it will not be recognized.

7.2. Network Connection

Q5: What should I do if the development board has no network connectivity, cannot be pinged, or fails to obtain an IP address?

A: Please troubleshoot in the following order:

1. **Physical Connection:** Confirm that the Ethernet cable is correctly connected to the development board (SoC Gigabit Ethernet port) and the router/switch.
2. **MAC Address Check:** Execute `print ethaddr` in the U-Boot command line to check if the MAC address is correctly set. The output should be in a format similar to `ethaddr=xx:xx:xx:xx:xx:xx`.
3. **Set MAC Address:** If `ethaddr` is empty or needs modification, please execute the following command (replace the example address with your valid MAC address):

```
=> env default -a
=> env set ethaddr aa:bb:cc:11:22:33
=> saveenv
=> reset
```

7.3. System Flashing and Partitioning

Q6: After flashing is complete, the system fails to boot or reports boot errors.

A: This is usually caused by legacy environment variables from an older version of U-Boot being incompatible with the new system. Please try the following steps to reset the environment variables after flashing the new Bootloader:

```
=> env default -a
=> saveenv
=> reset
```

Warning

`env default -a` will clear all manually configured environment variables you previously set (such as MAC address, boot arguments, etc.). Please proceed with caution and reconfigure them after execution.

Q7: Flashing prompts “partition does not exist” or “partition is too small” .

A: This indicates that the partition table on the target storage device (eMMC) is incorrect or missing. Please execute the following steps in order:

1. **Confirm Bootloader is functional:** Ensure that the correct `bootloader.bin` has been flashed and the system has booted.
2. **Clear old environment variables:** Execute `env default -a` and `saveenv` (see Q6).
3. **Re-partition:** Execute the standard partitioning command:

```
=> run gpt_partition
```

Once you see the `Writing GPT: success!` prompt, you can retry flashing the system image.

Q8: The default root partition size is insufficient. How do I adjust and re-partition?

A: You need to customize the partition table parameters and then recreate the partitions.

1. View current partition definition: `=> print partitions`
2. Edit the variable: `=> edit partitions`, or directly use `env set partitions "...[Your Partition Table String]..."` to set it.
3. Apply the new partition table: `=> run gpt_partition`

Q9: How do I confirm that the flashed system image (Bootloader/Kernel) has been successfully updated?

A: Observe the serial logs during system boot:

- **Bootloader:** Early in the U-Boot startup, a line of information will display the compilation date and time of U-Boot. Compare this time with the compilation time of the image you flashed.
- **Linux Kernel:** During kernel decompression and startup, it will also print its compilation timestamp. Comparing this time confirms whether the kernel image has been updated.

Q10: How do I partition an SD card or SATA hard drive?

A: Use the `gpt write` command, but specify the correct device identifier.

- **For on-board eMMC (mmc 0):** `gpt write mmc 0 $partitions`
- **For SD card (mmc 1):** First switch the device `mmc dev 1`, then execute `gpt write mmc 1 $partitions`.
- **For SATA devices:** Replace `mmc` with `sata`, e.g., `gpt write sata 0 $partitions`.